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Instruction Manual

Rotating Cutting Force Dynamometer (RCD)
Type 9170B...

9170B_012-021e-06.25

Foreword

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1. Introduction

Please take the time to thoroughly read this instruction manual. It will help you with the installation, maintenance, and use of this product.

To the extent permitted by law Kistler does not accept any liability if this instruction manual is not followed or products other than those listed under Components and Accessories in chapter 10.4 are used.

Kistler offers a wide range of products for use in measuring technology:

- Piezoelectric sensors for measuring force, torque, strain, pressure, acceleration, shock, vibration and acoustic-emission
- Strain gage sensor systems for measuring force and torque
- Piezoresistive pressure sensors and transmitters
- Signal conditioners, indicators and calibrators
- Electronic control and monitoring systems as well as software for specific measurement applications
- Data transmission modules (telemetry)

Kistler also develops and produces measuring solutions for the application fields engines, vehicles, manufacturing, plastics and biomechanics sectors.

Our product and application brochures will provide you with an overview of our product range. Detailed data sheets are available for almost all products.

If you need additional help beyond what can be found either on-line or in this manual, please contact Kistler's extensive support organization.

2. Important notes

Please do not fail to observe the following instructions: Your compliance is in the interest of your personal safety while working with the rotating cutting force dynamometer (RCD) and ensures a long, trouble-free operation of the device.

2.1 General

- Check all of the packaging of the device for transport damage. Report any damage to the transport company and the responsible Kistler representative
- Work with the device should be carried out only under the specified operating conditions; it should be protected against the effects of excessive dust and against mechanical strain (shock, vibration)
- High humidity, such as could lead to condensation with temperature changes, is to be avoided

2.2 For your safety

- This device was thoroughly checked and left the plant in an impeccable condition with respect to safety considerations. To maintain this condition and to ensure hazard-free operation, the user must observe the instructions and warning notes that are contained in this Instruction Manual and/or printed on the device
- Also observe local safety regulations governing the handling of mains-fed electrical and electronic devices
- If it is to be assumed that hazard-free operation is no longer possible, then the device is to be put out of operation and secured against being started up inadvertently.

It is to be assumed that hazard-free operation is no longer possible...

- if the device exhibits visible damage
- if the device no longer functions
- after prolonged storage under unfavorable conditions
- after heavy transport strains

If, in accordance with the above-mentioned features, hazard-free operation is no longer ensured, the device must be sent to the manufacturer's plant or the responsible representative for repair without delay.

2.3 Electromagnetic compatibility

The rotating dynamometer Type 9170B... and the LabAmp WL Receiver Type 5347A... are CE-compliant and meet the standards for electromagnetic compatibility (EMC) (see chapter 13).

2.4 Instructions on handling the devices

2.4.1 Dynamometer

The rotating dynamometer may be installed, operated and serviced only by persons who are familiar with the device and who are sufficiently qualified for their respective work.



Fig. 1: Pressure relive valve



Attention!

- Mount the dynamometer in accordance with regulations in the spindle of the machine tool
- The maximum speed – engraved on the rotor and/or specified in these instructions in Chapter 10 – may not be exceeded
- Only tools that have been approved by the tool manufacturer for the corresponding speed may be used
- The rotating cutting force dynamometer may be used only with the safety devices (protective doors, etc.) that are prescribed by the machine tool manufacturer
- Threaded connections on the dynamometer may **NOT** be loosened. The precision balancing would otherwise be lost. Only the manufacturer is permitted to dismount the dynamometer
- The pressure relief valve (Fig. 1) is for safety and must be kept free of obstructions and must not be covered.

The permitted load on the Rotating cutting force dynamometer – see Technical Data chapter 10 – may not be exceeded.

2.4.2 Electronics

- Observe local safety regulations governing the handling of mains-fed electrical and electronic devices.
- Energized parts could become exposed when the covers are opened or components are removed while the device is connected.
- Calibration, maintenance and repair tasks on energized, open devices should be avoided as much as possible. If however such tasks are unavoidable, they may be carried out only by a technician who is familiar with the associated hazards. The device must be disconnected from all voltage sources before any of these tasks is performed.

2.5 Tips for using the instruction manual

We generally recommend that you read the entire Instruction Manual. If, however, you are in a hurry and already have experience with Kistler Dynamometers, then you can limit your reading to the information you currently require (see chapters 4 and 5).
We have endeavored to structure these instructions in a clear manner and thus to facilitate direct access to the information you are looking for.

Please store this Instruction Manual in a safe place where it always ready at hand.
If these instructions become lost, please contact your Kistler Customer Service Office and you will receive a replacement without delay. All of the information and directives contained in these instructions may be changed at any time and without advance notification.

3. General system description

3.1 Purpose of the rotating Cutting Force Dynamometer (Inteded Use)

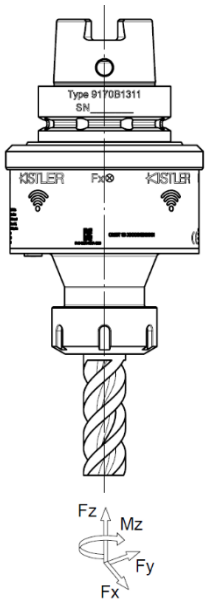


Fig. 2: RCD Type 9170B… with coordinate system

The rotating cutting force dynamometer (RCD) is used for the dynamic and quasi-static measurement of the three Cartesian force components F_x, F_y, F_z and the torque M_z on a rotating tool. Data transmission is accomplished with radio link.

The dynamometer is compact in construction, has considerable rigidity and thus a high natural frequency, whereby the usable frequency range is determined by the dynamometer, the attached adapter and tools, the machine tool and the radio system.

Rotating cutting force dynamometers are suitable for basic research and application investigations of production processes, particularly of milling and drilling processes.
In cases of critical tools and expensive workpieces, the device can also be used to monitor the process.

3.1.1 Application examples

The rotating cutting force dynamometer (RCD) can be used for a wide variety of measurement tasks in the measurement laboratory. These include, for example:

- Optimization of drilling, milling, reaming and similar tools
- Investigation of wear mechanisms
- Determination of tool life
- Investigation of cutting processes (interactions between tool cutting edge and workpiece)
- Process monitoring

Rotating measurement systems have the advantage over measurement platforms (i.e. stationary dynamometers) of measuring the cutting forces on the rotating tool independently of workpiece weight and workpiece size.

3.2 Components of the measurement system



Fig. 3: RCD measurement system

The rotating dynamometer system consists of:

- Dynamometer (rotor) with built-in charge amplifiers and radio electronics (type 9170B....)
- Receiver dongle with USB extension cable
- LabAmp WL Receiver (type 5347A…)

3.2.1 The rotating Dynamometer

- A 4-component sensor (F_x, F_y, F_z, M_z) is built into the Rotating Dynamometer Type 9170B... that is built in using a hollow tension screw under high preload between machine adapter and the tool adapter. This is the prerequisite for the transfer of the friction forces for the measurement of the shearing forces F_x and F_y and the torque M_z
- The dynamometer is corrosion-resistant and secured against the entry of splash water and coolant (IP67).
- For each of the 4 components (F_x, F_y, F_z, M_z) a miniature charge amplifier is built into the rotor (dynamometer). The power supply to the amplifiers is provided by a battery. Reset/operate and range switching are transmitted via radio link.
- The output signal of the charge amplifier is digitized with 16-bit resolution and transmitted by radio to the receiver side.

3.2.2 The Receiver Dongle

The receiver dongle is the receiving and transmitting unit on the stationary side of the measuring system. The receiver dongle contains the transmitting and receiving antenna.
The receiving antenna receives the data signals transmitted by the rotor and forwards them to the LabAmp WL Receiver.
The dongle should be placed as close as possible to the rotor. A USB extension cable of max. 3 m can be used for this purpose. Longer cables must not be used.

3.2.3 The LabAmp WL Receiver

The WL Receiver is the central device of the system on the receiver side. The data stream received from the rotor is routed from the receiver dongle to the WL Receiver.
The data is then processed internally and scaled for the output side. The data is output through a data stream (Ethernet connection) or via analogue outputs (+/- 10 V).
Low-pass filters can be set in the WL receiver via software such as the PTS app.

3.2.4 The Battery

3.2.4.1 Charging

The battery runtime (active measurement) is at least 5 hours. Afterwards, the RCD with built-in battery can be charged via the attached plug.



Only use the original accessories to charge the battery!

Connect the country-specific mains adapter to the charger. Press the jack to replace the battery.

Only charge the RCD at an ambient temperature between 0 and 45 °C. Use the original charger and the original charging cables. Charging outside the specifications can seriously and irreversibly affect the performance and service life of the rechargeable battery!

1. Plug the charger with the country-specific cable into a supply voltage of 90 ... 264 VAC (~47 ... 63Hz).
2. The charging process starts immediately when the charging cable is connected.
3. A complete charging process takes a maximum of 3 hours.
4. As soon as the RCD battery is fully charged, the cable can be disconnected.

3.2.4.2 Storage

Self-discharge
Typical self-discharge in the first year at a temperature of 20 °C is less than 5 %, with a state of charge between 15 and 50 %.

Storage temperature
The recommended storage temperature is between –20 °C to 45 °C out of direct sunlight in a dry, clean, well-ventilated place.



Storage outside these temperatures can lead to higher self-discharge, lower performance and swelling of the battery cells.

Minimum state of charge during storage
A minimum state of charge of 25% at the end of storage is necessary to prevent deep discharge.



Deep discharge can have a serious and irreversible effect on the performance and life of the battery! If the battery voltage is low after storage, the battery must be charged as soon as possible to avoid the voltage of one or more cells falling below a level where the cell may be damaged. To do this, "wake up" the battery by connecting the charger supplied. If the unit cannot be switched on after charging or if the state of charge is permanently below the permitted level of 25%, please contact Kistler Service.

For storage of more than 3 months to less than 1 year, proceed as follows:

- Regularly check the battery voltage at least every 6 months.
- If necessary: Charge the battery up to 50 % of its capacity.

Use of the RCD after a long storage period
After a long storage period, the following procedure is recommended:

- Completely discharge the battery by running the RCD in operating mode (measuring mode).
- Preferably run a complete charge/discharge cycle. Otherwise it may take 2 to 3 cycles until the maximum capacity is available again.
- Fully charge the battery

3.2.4.3 Safety instructions



Warning!

- **Do not attempt to remove the battery (battery can only be replaced by Kistler)**
- **Do not dismantle the RCD**
- **Do not expose to heat or flames**
- **Do not short circuit**
- **Do not drop or hit forcefully**
- **Charge only with the original charger**

3.2.4.4 Transport of RCDs

Legal framework



Due to the installed lithium-ion battery, the RCDs must be shipped as dangerous goods. UN number **UN 3481 applies.**

International and national transport regulations for air, sea, road and rail transport (ICAO/IATA, IMDG, ADR, RID...) are legally binding. Personnel preparing the RCD for transport must comply with all relevant requirements, including but not limited to special provisions, packaging requirements, labelling and training.

Overview
Personnel preparing the RCD for transport must carefully prescribe the applicable measures and instructions. Further information can be found in the relevant official documentation:
http://www.unece.org/trans/danger/danger.html

Among other things, the following must be taken into account:

- The mode of transport: air (IATA), sea (IMDG), road (ADR) or rail (RID)
- The country of origin and destination
- The applicable UN code UN 3481 and the associated labelling "Lithium-ion battery packed in equipment".
- The UN test certification status of the battery: UN38.3

Air transport



For the individual RCD, **PI 967, UN 3481, Class 9, Part 1** applies

3.2.5 Watchdogs

A watchdog is a mechanism for detecting and correcting firmware malfunctions. The sensor has 2 watchdogs:

- System watchdog
- Charging cable watchdog

The sensor is automatically restarted when one of its watchdogs triggers (detects a problem).
This is a precautionary measure in case the sensor no longer works as expected.

3.2.5.1 System watchdog

The firmware reloads the software watchdog at regular intervals. If the watchdog is not loaded for 60 seconds, the hardware triggers a reset, which restarts the sensor (and disconnects the radio link).

3.2.5.2 Charging cable watchdog

The charging cable watchdog is a software service that runs in the sensor firmware. It checks every second whether the charging cable is connected.
This watchdog can be used to restart the sensor if it no longer responds via radio (cannot be found or acts strangely).

To restart the sensor and break the radio connection, simply connect the live charging cable to the sensor.
However, there are two limitations to unintentional resets:

- The sensor must have been in operation for at least 30 seconds. This means that every connection/disconnection of the charging cable is ignored by the watchdog during this start-up time).
- The charging cable must have been disconnected for at least 10 seconds. This means that if the cable was connected and you want to restart the watchdog, you must disconnect the cable, wait 10 seconds and only then reconnect the cable.

3.3 Options and accessories

3.3.1 Spindle adapter

The rotating dynamometer (RCD) is equipped with the spindle adapter that you have specified. This is part of the total structure and cannot be replaced. Because of the high maximum speeds for which the RCD Type 9170B… is designed, the option of replaceable spindle adapters must be dispensed with, as the requirements with respect to the balance quality of the rotors is not ensured. An unbalance of the rotor can cause severe damage, not only to the machine spindle but also to the RCD.

Conversion to a different spindle adapter is possible to only a limited extent and must in any event be carried out at the manufacturer's plant. Contact the Kistler representative in the event that a conversion to a different spindle adapter should become necessary.



Threaded connections on the spindle adapter and on the rotor may not be loosened or dismantled under any circumstances.

3.3.2 ER collets

The Rotating Dynamometer Type 9170B... is equipped with a tool holder based on ER collets. DIN 6499-B32 collets are to be used exclusively. Collets that deviate from this standard could damage the dynamometer. Various manufacturers offer these collets. As an alternative, they can also be ordered directly from Kistler. Ultra-precision collets with the clamping range of 1 ... 22 mm are offered under the designation Type 9169A…. The assembly of the collet is described in chapter 4.2.1.

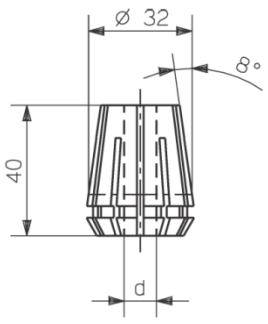


Fig. 4: DIN 6499-B32 collet chuck

3.3.3 Clamping nuts and sealing disks for internal cooling

The Dynamometer Type 9170B… enables the utilization of internal coolant of tools up to 70 bar. This means, however, that the corresponding clamping nuts and sealing disks have to be used, which can be obtained from specialist dealers. We recommend the use of the clamping nuts Hi-Q/ERC 32 (DIN 6499) from the RegoFix Co. with the appropriate sealing disks.

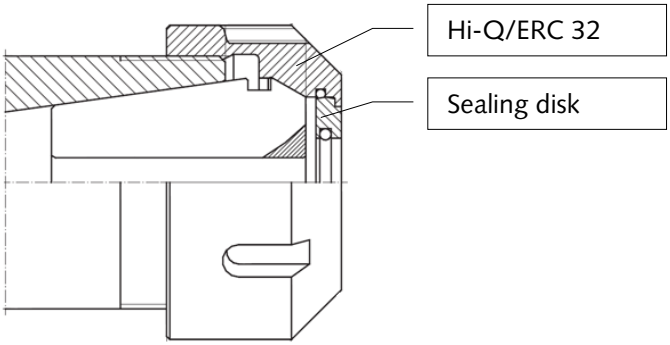


Fig. 5: Tool holder with Hi-Q/ERC 32[1] clamping nut and sealing disk

3.3.4 Data acquisition software

The Kistler PTS App type 2935A is a universal and easy-to-use software for data acquisition and evaluation. The configuration and control of the WL Receiver Type 5347A4 is integrated in this software.

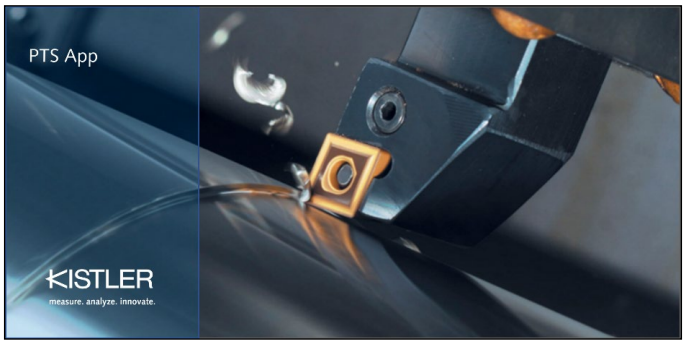


Fig. 6: Home screen of PTS App type 2935A

[1] Hi-Q is a registered trademark of Rego-Fix AG

4. Installation and commissioning

4.1 Important instructions

The RCD Type 9170B… is a precision instrument that requires careful handling. The following instructions should be followed for that reason:

- The dynamometer has been fine-balanced in order for it to be able to be used for up to 16 000 rotations per minute for force and moment measurements (see chapter 10). Depending on the selected adapter, there may be further restrictions on the rotation speed. The binding permissible speed is engraved on the housing of the RCD.
- Incorrect handling of the dynamometer has an immediate influence on the balance quality and thus on the quality of the measurement.



Do not allow the dynamometer to fall and do not subject it to any hard blows! The maximum force of a shock of this type could exceed the load range of the device and cause lasting deformations.

4.2 Mounting the tool

The tool is mounted prior to installation of the Rotating Dynamometer Type 9170B… on the spindle. The clamping range of the collet DIN 6499-B32 (Type 9169A...) must match the nominal diameter of the tool.

4.2.1 Mounting the collet

First, fit the notch of the collet DIN 6499-B32 in the eccentric ring of the clamping nut at the marked position. Tilt the collet chuck in the opposite direction until it audibly engages. Insert tool. Screw the clamping nut with the engaged collet on the thread of the holder. We recommend tightening the clamping nut with a torque wrench.

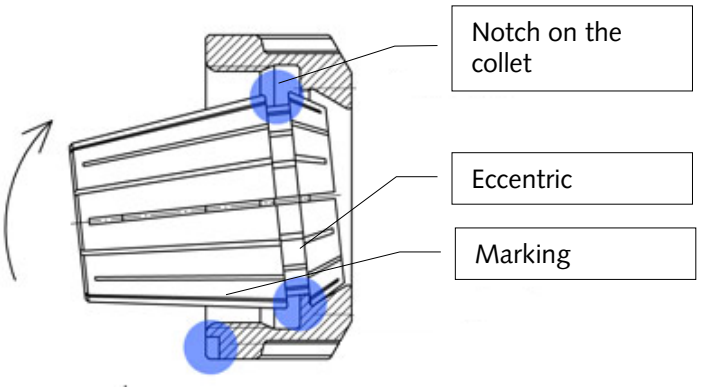


Fig. 7: Assembling the collet

The clamping nut Hi-Q/ER 32 or Hi-Q/ERC 32 (chapter 3.3.3) is to be tightened carefully with the help of the hook wrench included in the scope of delivery. The following tightening torques apply:

Tool diameter	Tightening torque [Nm]
2 … 2.5	24
3 … 22	136

4.2.2 Dismounting the collet

After unscrewing from the holder, press on the front side of the collet while at the same time removing the collet from the engagement of the nut by applying transverse pressure to the rear part, across from the marking.

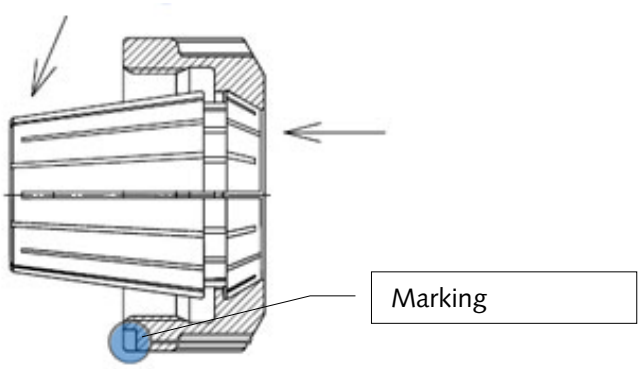


Fig. 8: Dismounting the collet

4.2.3 Axial stop

A mechanical stop is installed in the rotor in a centered position that can be shifted in axial direction using an Allen wrench. This stop ensures that the tool does not shift axially during processing. The tool is to be introduced into the collet chuck up to the stop.

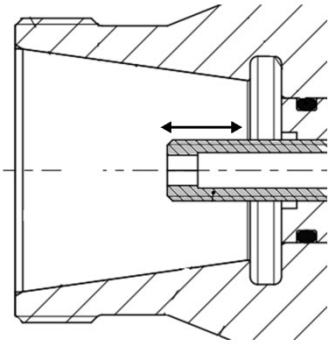


Fig. 9: Movable mechanical stop

4.3 Mounting the rotor (Dynamometer)



- Please observe the respective information in the manual of the machine tool manufacturer before the dynamometer is inserted in the spindle.
- The rotor may **only** be inserted **manually** into the spindle without explicit approval from Kistler and may not be changed with the automatic tool changer.

- The dynamometer is installed in the machine tool spindle using the conical spindle adapter. The mounting in the spindle is accomplished in accordance with the way that tools are mounted in the spindle.
- Once the installation has been completed, perform a visual check for dynamometer concentricity while the spindle slowly rotates.

4.4 Installation and commissioning of the receiver

Set up the receiver:

- The receiver dongle has to be connected to one of the USB ports on the back of the WL receiver 5347A4. For a better positioning of the receiver dongle, a USB extension cable can be used.
- Connect the power supply unit to the WL Receiver 5347A4
- Connect the WL Receiver 5347A4 to a Windows PC with an Ethernet cable.



It is mandatory to use the Sync In connection. The Sync Out connection is not active!



Fig. 10: Sync In connection of WL Receiver 5347A4

- After the cables have been connected correctly, the unit can be switched on.
- After the unit has started up (yellow flashing of the LEDs), the unit switches to the operating mode (blue flashing of the LEDs).

The software connection and set-up of the unit is described in the instruction manual of the software used, for example the PTS App Type 2935A.

4.5 Using the remote control interface

When the *Remote Control* connector is used, the measurement chain is controlled/triggered.

This control can be executed either via a programmable logic controller (PLC) or via a switch, e.g. type 2233B (proximity switch).

See chapter 10.2.2 for more details on the remote control interface.

4.6 Ranges of rotational speed to be avoided



Particularly with heavy tools or ones with long projecting lengths, it is imperative that a frequency analysis be carried out prior to operating. The tooth passing frequency may not under any circumstances fall in the range of the natural frequencies of the system or of individual components of the tool machine.

Chapter 6.4 is to be read through in detail and the instructions contained there are to be followed prior to commissioning.

4.7 Removing the rotating Dynamometer



The rotor may **only** be removed from the spindle **manually** without explicit approval from Kistler and may not be inserted with the automatic tool changer.

The measuring chain is to be switched off before the rotor is removed.

5. Forces and loads

5.1 Relationship of the effective forces

5.1.1 Cutting force and normal cutting force



The coordinate system of a rotating dynamometer rotates along with the tool. This is to be taken into account when interpreting the forces.

When measuring with RCD Type 9170B, the radial forces F_x and F_y, the axial force F_z and the torque M_z are acquired. The alignment of the radial forces $F_{x,RCD}$ and $F_{y,RCD}$ do not match this cutting edge, depending on the respective alignment of the cutting edge. The cutting force F_c and the normal cutting force F_{cN} can be calculated with the help of the active force F_a (which is made up of the combined measured forces $F_{x,RCD}$ and $F_{y,RCD}$) and of the measured torque M_z.

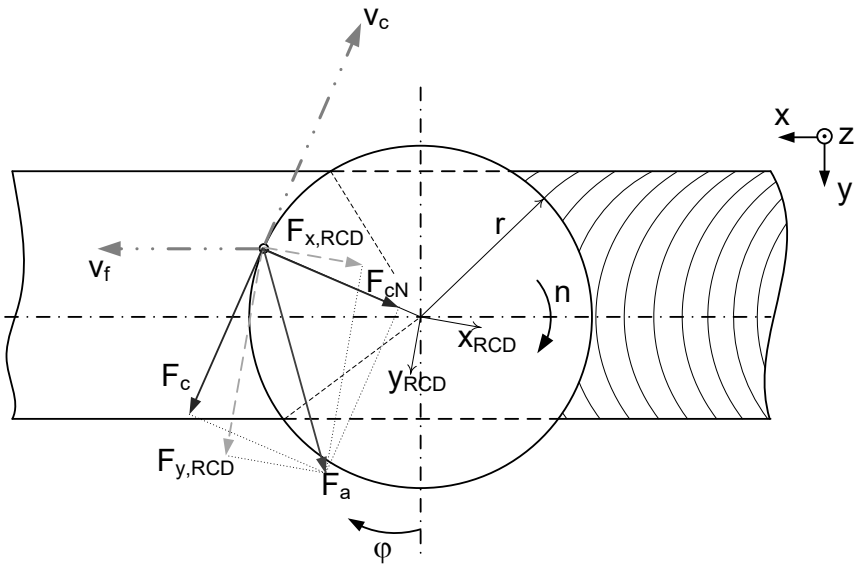


Fig. 11: Forces during milling acquired with an RCD

$$F_a = \sqrt{{F_{x,RCD}}^2 + {F_{y,RCD}}^2} = \sqrt{{F_c}^2 + {F_{cN}}^2}$$

$$F_c = \frac{M_z}{r}$$

$$F_{cN} = \sqrt{{F_a}^2 - {F_c}^2}$$

5.1.2 Relationship of the forces measured with an RCD and with a stationary Dynamometer

In contrast to stationary dynamometers, the coordinate system rotates along with the rotating dynamometers. For this reason, the measurement signals of an RCD ($F_{x,RCD}$ and $F_{y,RCD}$) cannot be compared directly with the measurement signals of a stationary dynamometer. To accomplish this, the angle of rotation φ must be known. With it, the feed force F_f and the normal feed force F_{fN} can be calculated.

The RCD Type 9170B is not in a position to acquire and output the position of the rotating angle φ. Machine controls of the more recent machine tools enable angle signals to be output with little effort. Please contact the manufacturer of your machine tool for this purpose.

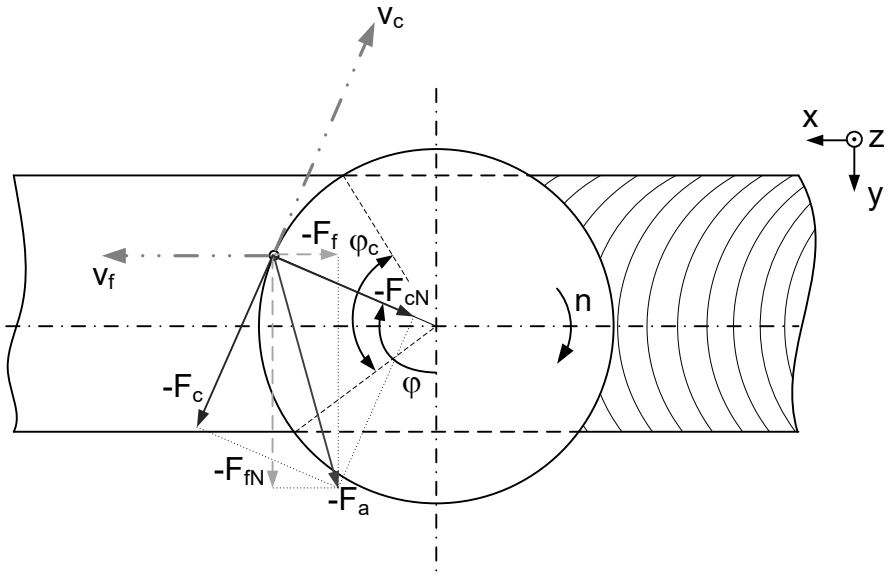


Fig. 12: Comparison of forces during milling

$$|F_{x,stat.Dyno}(\varphi)| = |F_f(\varphi)| = |-F_c(\varphi)\cos\varphi - F_{cN}(\varphi)\sin\varphi|$$

$$|F_{y,stat.Dyno}(\varphi)| = |F_{fN}(\varphi)| = |+F_c(\varphi)\sin\varphi - F_{cN}(\varphi)\cos\varphi|$$

$$|F_c(\varphi)| = |F_f(\varphi)\cos\varphi + F_{fN}(\varphi)\sin\varphi|$$

$$|F_{cN}(\varphi)| = |F_f(\varphi)\sin\varphi - F_{fN}(\varphi)\cos\varphi|$$

5.1.3 Direct comparison of the forces measured with an RCD and stationary Dynamometer

Because of the rotating coordinates, no direct comparison of the forces F_x and F_y by RCD and by stationary dynamometer is possible (see chapter 5.1.2). Only the active force is subject to direct comparison.

$$F_a = \sqrt{F_{x,RCD}{}^2 + F_{y,RCD}{}^2} = \sqrt{F_c{}^2 + F_{cN}{}^2}$$

$$F_a = \sqrt{F_{x,stat.Dyno}{}^2 + F_{y,stat.Dyno}{}^2}$$

5.2 Determining the maximum loading capacity of the Dynamometer

The RCD is subjected to torsion and bending strain during the cutting process. Torsion and bending tension may not exceed certain limits in order to ensure that the piezo-electric elements of the sensor are not over-strained.

A description is given below as to how the cutting forces and thus also the dynamometer load are calculated.

The measurement of individual cutting force components F_c, F_f, F_p can be carried out only with a single-tooth tool.



A rotating dynamometer always measures the total of all effective forces. If several cutting edges are currently in use at the same time, then only the total of all of the forces effective in one direction can be measured.

5.2.1 Calculative determination of the Cutting Force F_C

The fundamental dimension for the calculation of the cutting force F_c and of the normal cutting force F_{cN} is the chip thickness h.

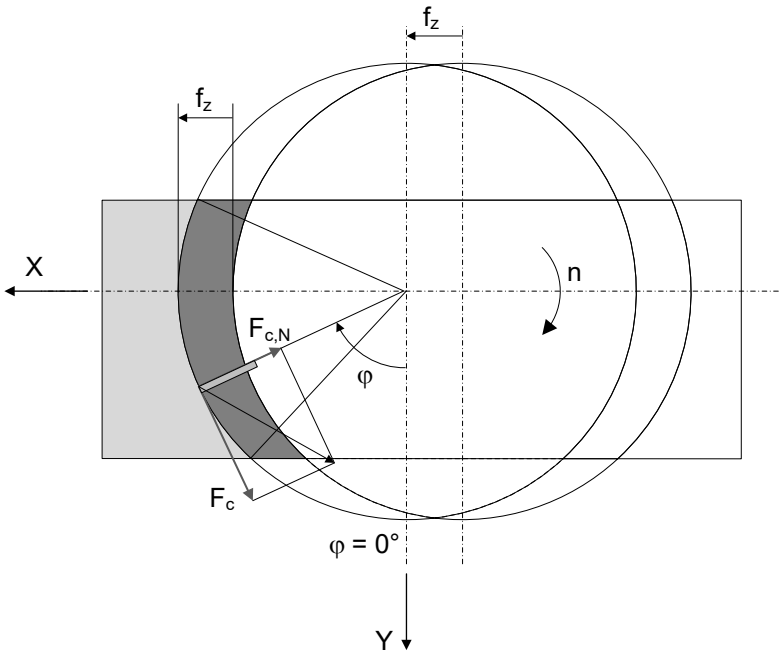


Fig. 13: Fundamental dimensions during milling

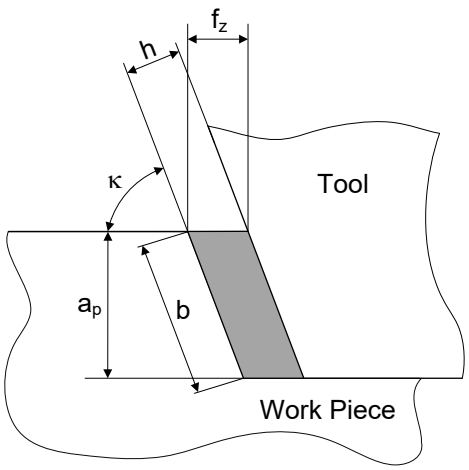


Fig. 14: Chip geometry

This chip thickness h is a function of the immersion angle, φ, by which f_z is the feed rate per tooth and κ is the setting angle.

$$h(\varphi) = f_z \sin\kappa \sin\varphi$$

The chip width b is made up of the axial infeed a_p and the tool cutting edge angle κ together.

$$b = \frac{a_p}{\sin \kappa}$$

According to Kienzle, the cutting force is based in simplified terms on the chip width b, the specific cutting force $k_{i1,1}$ (for benchmarks, see table 1) and the chip thickness h and the tangent value of the angle of inclination m (for benchmarks, see table 1). Various influences (wear, coolant, etc.) are not taken into account thereby. Furthermore, the rule applies that only one cutting edge is currently in use.

$$F_i(\varphi) = b \cdot k_{i1,1} \cdot h(\varphi)^{1-m}$$

or simplified

$$F_c = b \cdot k_{c1,1} \cdot h^{1-m} \quad \text{with} \quad i = c$$

The normal cutting force can be calculated the same way, for which x corresponds to the tangent value of the angle of inclination of the specific normal cutting force

$$F_{cN} = b \cdot k_{cN1,1} \cdot h^{1-x} \quad \text{with} \quad i = cN$$

The active force follows from the cutting force and the normal cutting force.

$$F_a = \sqrt{{F_c}^2 + {F_{cN}}^2}$$

The specific cutting force $k_{c1,1}$ and the specific normal cutting force $k_{cN1,1}$ are specified in table 1 for several materials and cutting parameters:

Table 1: Median specific cutting forces and normal cutting forces in metal-cutting technology (according to *W. Degner, H. Lutze, E. Smejkal: Chip Machining*)

Material		Cutting Force F_c (1–m)	$k_{c1.1}$ *N/mm²*	Normal Cutting Force F_{cN} (1-x)	$k_{cN1.1}$ *N/mm²*	Passive Force F_p (1-y)	$k_{p1.1}$ *N/mm²*
S355, E295 (St50)	1.0050	0.74	1990	0.299	351	0.51	274
E360 (St70-2)	1.0070	0.70	2260	0.384	364	0.51	311
C15	1.0401	0.78	1820	0.199	333	0.20	260
C35	1.0501	0.73	1516	0.20	321	0.46	259
C45E	1.0503	0.86	2200	0.325	343	0.52	263
C60E	1.1221	0.82	2130	0.288	347	0.59	250
42CrMo4	1.7225	0.74	2500	0.329	334	0.52	271
16MnCr5	1.7131	0.74	2100	0.30	391	0.54	324
20MnCr5	1.7147	0.75	2140	0.32	337	0.48	246
100Cr6	1.2067	0.61	1410	0.14	318	0.47	362
55NiCrMoV6	1.2713	0.79	1595	0.21	269	0.34	198
50CrV4	1.8159	0.74	2220	0.23	317	0.61	315
X10CrNiMoNb18 10	1.4580	0.82	2550	0.26	181	0.41	173
EN-GJL-200 (GG20)	0.6020	0.75	1020	0.301	240	0.54	178
EN-GJL-250 (GG30)	0.6030	0.74	1470	0.09	170	0.30	164
EN-GJS-600-3 (GGG60)	0.7060	0.83	1480	0.240	290	0.57	240
G-AlMg4SiMn	3.3251	0.80	487	-0.08	20	0.25	32

Measurement Conditions:
- Cutting Velocity v_c = 100 m/min
- Tool: Solid Carbide
- Setting angle κ = 70°;
- Tool angle ε = 90°;
- Rake angle λ = 0°;
- Corner radius r_n = 0.8 mm;
- Clearance angle α = 5°
- Rake angle χ = 6° (Steel), χ = 15° (Aluminum), χ = 2° (Cast Iron)

Calculation example:
Determination of the cutting force and the normal cutting force during milling (calculative).

- Basic material to be processed: 18 CrNi6
- Setting angle κ : 90°
- Feed rate per tooth f_z: 0,05 mm
- Axial infeed a_p: 5 mm
- Angle of immersion: 180°
- Only one cutting edge in use
- The influence of the helix angle is neglected

The greatest force occurs with the greatest chip thickness h_{max}, i.e. with $\varphi = 90°$

$$h_{max} = f_z \sin\kappa \sin\varphi = f_z = 0.05mm$$

$$b = \frac{a_p}{\sin\kappa} = \frac{5}{\sin(90°)} = 5mm$$

$$k_{c1.1} = 2260\ \frac{N}{mm^2};\ (1-m) = 0.7 \quad \text{(from Table1)}$$

$$F_{c,max} = b \cdot k_{i1,1} \cdot {h_{max}}^{1-m} = 5 \cdot 2260 \cdot 0.05^{0.7}$$

$$F_{c,max} = 5 \cdot 2260 \cdot 0.05^{0.7} = 1387N$$



If several cutting edges are in use simultaneously, then the load on the measuring element can be several times greater than with a single engaged cutting edge, as it is always loaded with the total of all of the forces.

5.2.2 Determining the Dynamometer load

Several forces have an effect on the RCD Type 9170B during processing. The important thing here is to ensure that the measuring device does not become overloaded, thus necessitating costly repairs. To calculate bending moments, it is the dimension az, i.e. the distance between the tool tip and the shoulder of the rotor (see Fig. 15) that is of particular significance.

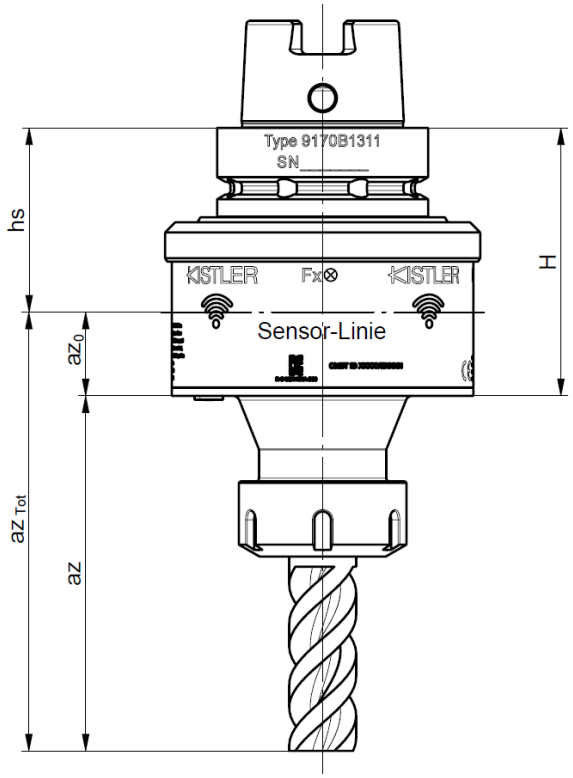


Fig. 15: Distance h und az on the RCD Type 9170B

5.2.2.1 Basic procedure for determining the Dynamometer load

Neither the load limit of the sensor in the interior of the RCD Type 9170B nor the maximum load of the spindle adapter may be exceeded during the machining. This is taken into account for the calibration of the Type 9170B; accordingly, there is a difference between the nominal measurement range and the calibration range (chapter 10.1)



Particularly with larger spindle adapters, care must be taken to ensure that the sensor does not become overloaded due to the forces in effect.

5.2.2.2 Dynamometer load with utilization of planar-fitting spindle adapters

In the case of spindle adapters with planar fit (e.g. HSK or Capto adapters), the effective lifting force and thus the bending moment may not exceed a certain value in order not to lose the planar fit. This planar fit becomes lost when the shear force vector F_{xry} in Fig. 16 with a known cantilever length az lies to the right of the regressive load curve of the corresponding machine adapter (–>Reading example: chapter 5.2.2.4).

Table 2: The following key figures form the basis of the planar fit (according to *Manfred Weck, Christian Brecher: Werkzeugmaschinen, Maschinenarten und Anwendungs-bereiche*)

Machine adapter	Clamping force	Lifting moment of the planar fit
HSK-A40	6.8 kN	140 N·m
HSK-A50	11 kN	230 N·m
HSK-A63	18 kN	450 N·m

The following picture shows the load limits of the dynamometer (red line with grey highlighted area) and the spindle adapter, which must not be exceeded.

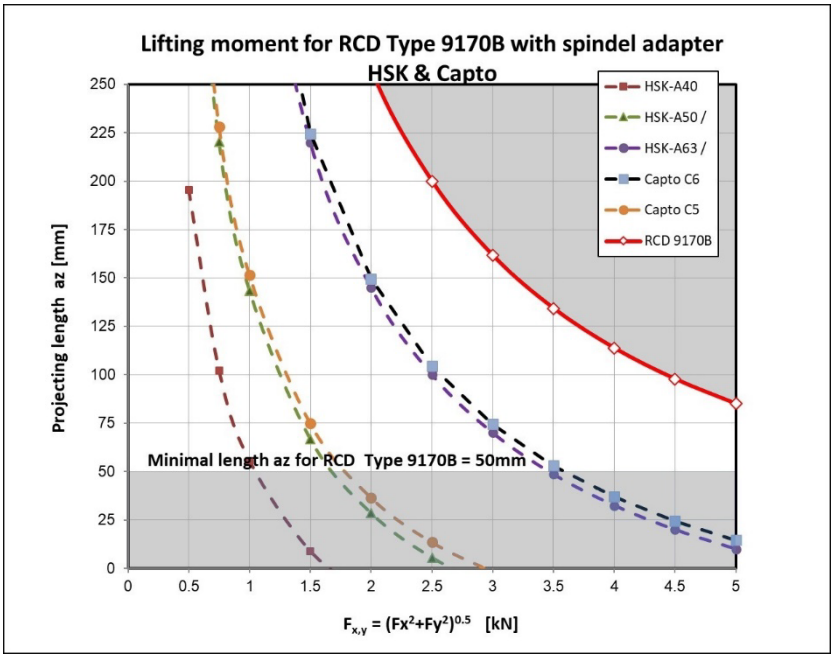


Fig. 16: Load diagram $F_{x,y}$

If the shear force vector $F_{x,y} = ((F_c^2 + F_{cN}^2)^{0.5})$ is known, then the maximum axial force F_z can be read from Fig. 17 These may also not be exceeded.

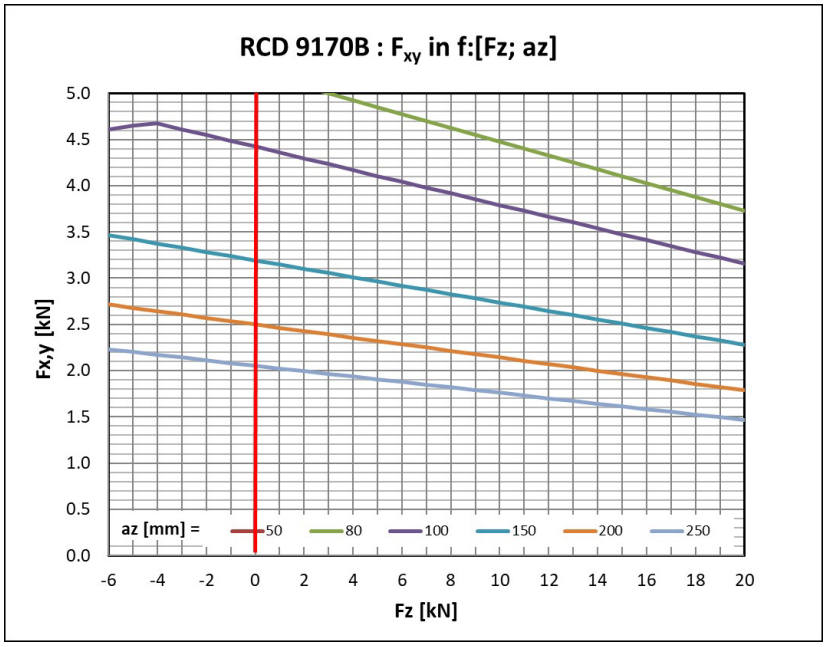


Fig. 17: Load diagram F_z

5.2.2.3 Dynamometer load with the utilization of taper shank machine adapters

The following picture shows the load limits of the dynamometer (red line with grey highlighted area) and the spindle adapter, which must not be exceeded.

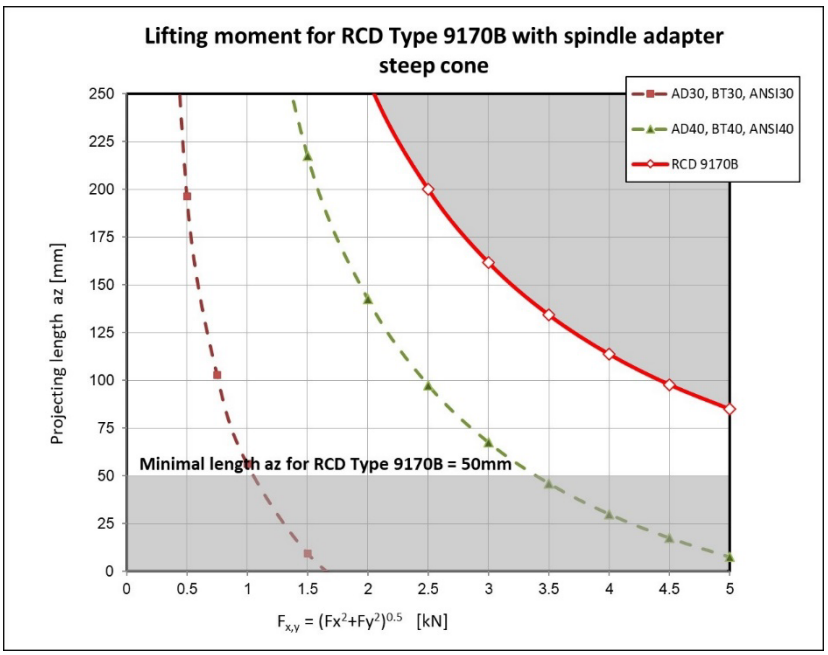


Fig. 18: Load diagram $F_{x,y}$

If the shear force vector $F_{x,y} = ((F_c^2 + F_{cN}^2)^{0.5})$ is known, then the maximum axial force F_z can be read from Fig. 17. These may also not be exceeded.

5.2.2.4 Reading example

- Machine adapter used: HSK-A63
- Cantilever length az: 150 mm

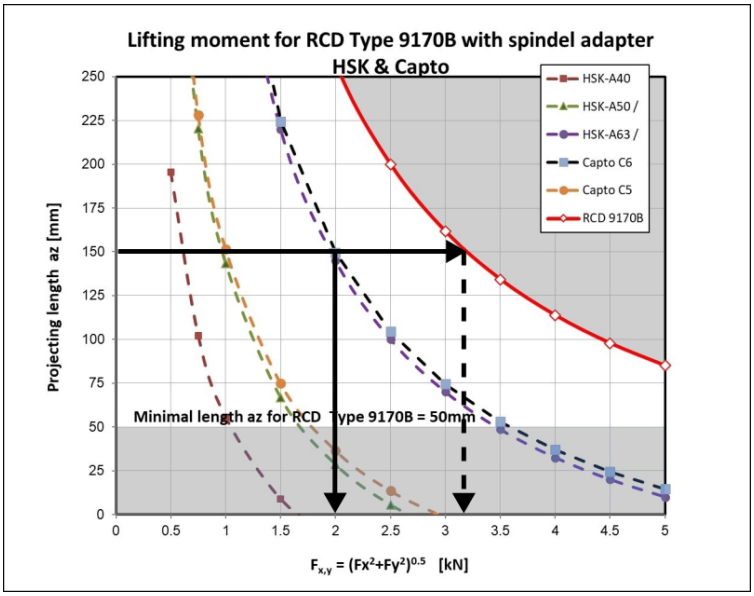


Fig. 19: Shear force vector $F_{x,y}$ for az = 150 mm

Result: If az = 150 mm, the planar fit would not be lost before the lifting force vector reaches $F_{x,y}$ ≥2.0 kN. The sensor would also not be overloaded as the operating point lies to the right of the regressive red curve –> $F_{x,y}$ ≤3.2 kN.

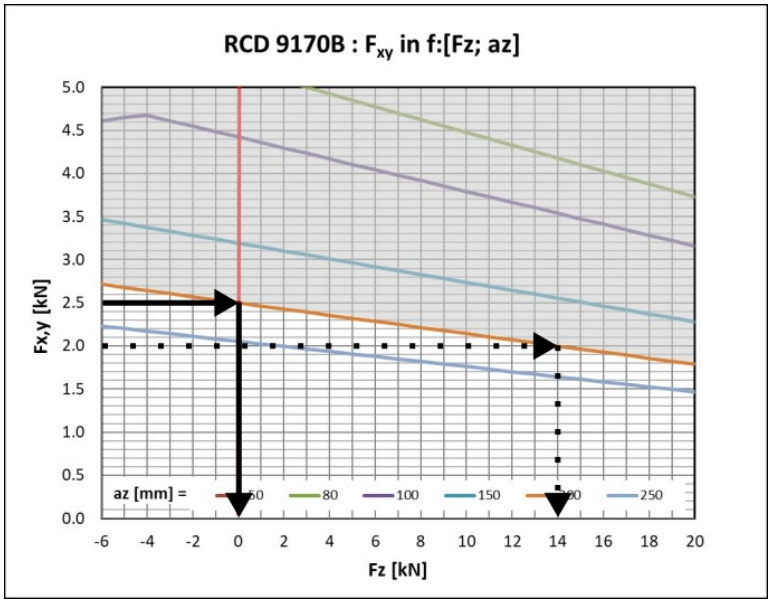


Fig. 20: Axial force F_z

Result: The RCD is overloaded when $F_{x,y}$ >2.5 kN. The axial force is not permitted to exist when $F_{x,y}$ = 2.5 kN (F_z = 0 kN). It would make sense then to change the cutting parameters to reduce the shear force vector $F_{x,y}$. Example: Reduction of $F_{x,y}$ to 2 kN –> max. permissible axial force F_z = 14 kN.

5.2.2.5 Torque load

The maximum torque load of the system is dependent primarily on the friction between tool and collet chuck, i.e. on the shaft diameter of the tool used (blue line in Fig. 21). The measurement range of the RCD is limited to 100 N·m (red line in Fig. 21). In addition, it is also dependent on the spindle adapter used. In particular, spindle adapters without drive slots (e.g. HSK-E…) exhibit extremely reduced torque ranges to some extent (Fig. 22). This circumstance is taken into account for the calibration of Type 9170B.

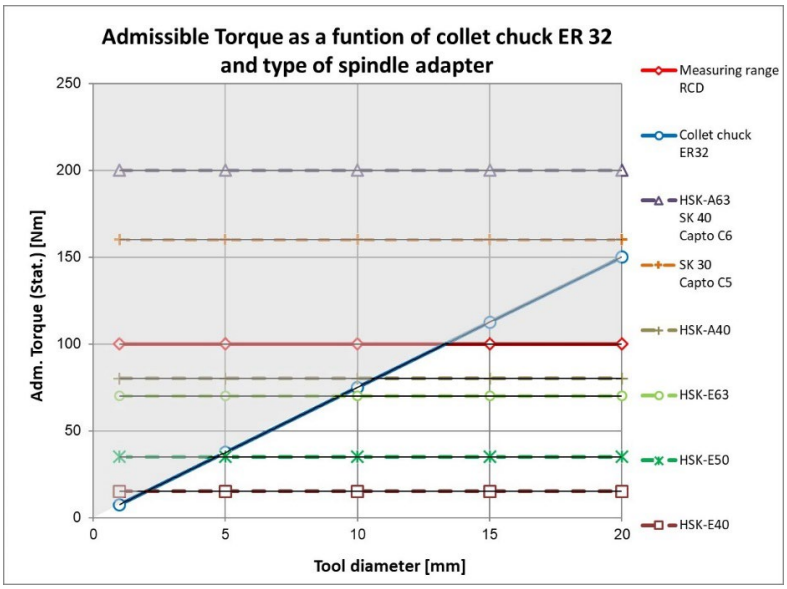


Fig. 21: Torque load diagram

5.2.3 Avoiding overloading in everyday practice

In order to avoid expensive repairs of the measurement system, it is important exercise caution with the first tests, particularly in cases where the respective load either cannot be estimated or can only be very poorly estimated beforehand.

Basically there are two possibilities to choose from:
1. Reduction of the chip width b
2. Reduction of the chip thickness h

The first is accomplished by reducing the axial infeed, the second by reducing the feed rate.

The reduction of the chip width brings about, in simplified terms, linear behavior, while the reduction of the chip thickness brings about regressive behavior.

The tables in chapter 10.1.1 are to be applied to the measured values in accordance with the reading example in chapter 5.2.2.4 and are to be checked to ensure that the loads that occur do not overload either the machine adapter or the sensor.

6. Operation

6.1 Operating and display elements

The RCD has no status displays. All information about the status of the unit is provided by the LEDs of the WL receiver and by the status information in the software.

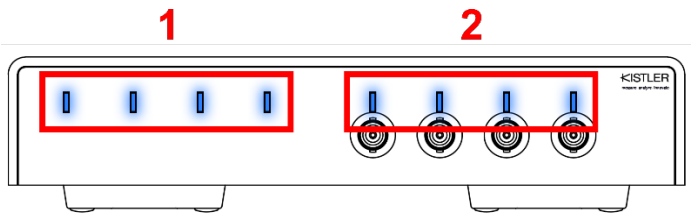


Fig. 22: Group 1 are input LEDs and group 2 are output LEDs

LED status		Operating Status	
		RCD input LEDs	Output LEDs
	no light	No RCD connected, unit switched off	-
	blue	RCD connected, battery ok, no overload	Measurement in progress
	blue flashing	-	Reset
	yellow	RCD connected, battery < 20%	-
	red	RCD connected, Overload	Overload
	red flashing	RCD connected, battery < 10%	-

6.2 Range selection

The unit has four measuring ranges for each measuring channel. Depending on the configuration of the adapters, larger measuring ranges may be deactivated because they cannot be used with the adapters. The usable measuring ranges are read out by the measuring or configuration software used and made available to the user for selection. For a detailed description, see the instructions in the respective instruction manual of the software, such as the PTS App.

Chapter 10.1.1 shows the measuring ranges depending on the adapters.

6.3 Usable frequency range of the mounted Dynamometer

The natural frequencies of the Rotating dynamometer are high in all directions due to the preload of the sensor, the very rigid mechanical construction and the small size of the device (see chapter 10.1). If however the dynamometer is installed in a machine tool spindle, then the natural frequencies of the machine spindle apply additionally.

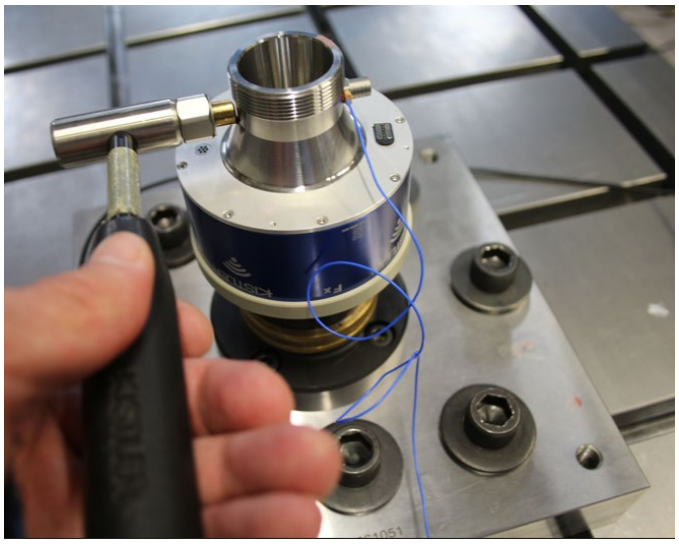


Fig. 23: Natural frequency measurement in **X/Y direction** of the RCD Type 9170B

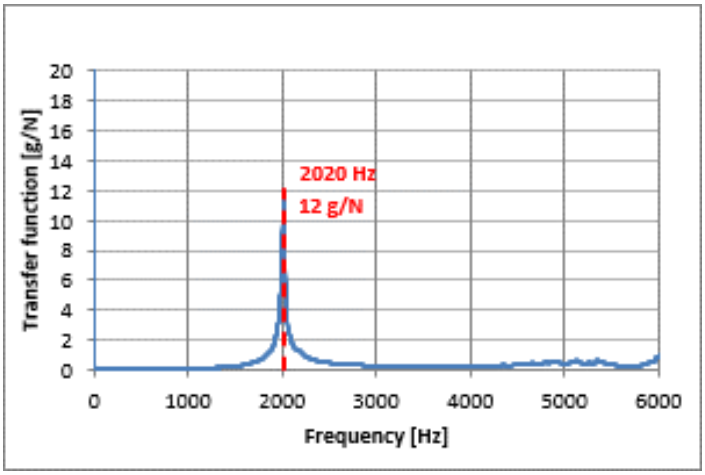


Fig. 24: Natural frequency in **X/Y direction** of the RCD Type 9170B, without tool

In order to avoid errors in the signal amplitudes, measurement must be carried out considerably below the natural frequency of the measurement setup. The following rule of thumb applies:

- For excess amplitude peaks <5 %, the excitation frequency (also referred to as the tooth passing frequency) may not amount to more than a maximum of $f_n/5$ during machining.
- For excess amplitude peaks <10 %, the excitation frequency may not amount to more than a maximum of $f_n/3$ during machining.

$$f_{tooth\ passing} = \frac{n}{60} \cdot N$$

n: Speed [min^{-1}]
N: Number of cutting edges of the milling cutter

Ideally, the RCD clamped in the machine spindle is excited on the attached tool by means of a light blow with an impact hammer and the transfer function is recorded. The natural frequency of the structure can be determined by this procedure.

One simple possibility – although one associated with some uncertainty – is to excite the structure with a lightweight hammer in a particular direction, e.g. in X direction. The (X) force signal is recorded on an oscilloscope. The reciprocal value of the period duration of the dying signal corresponds to the natural frequency of the dynamometer-tool-spindle-machine system, meaning also that the natural frequencies of the machine tool structure, of the spindle, of the tool, etc. are visible in the signal.

6.4 Dynamic behavior of the RCD



Particularly with heavy tools or ones with long projecting lengths, it is imperative that a frequency analysis be carried out prior to operating.

The frequency analysis of the structure is exceptionally important for preventing damage to the RCD, tool, workpiece and the machine tool. The RCD is excited periodically by the interrupted cuts that arise. If this excitation lies in the vicinity of or even on a natural frequency of a system component (RCD, spindle, machine structure, etc.), then the RCD may resonate and suffer severe damage. This behavior is additionally intensified by heavy tools or ones with long projecting lengths.

It is imperative that the frequency analysis be performed when:

- the weight of the tool is greater than 200 g
- the projecting length a_z (see chapter 5.2.2.2) is greater than 125 mm

6.4.1 Procedure for determination of the natural frequencies

1. Installation of the RCD and of the entire measuring chain in accordance with chapter 4, whereby the tool must be mounted beforehand in the rotor.
2. Switch on the measuring chain.
3. Start the measurement and hammering of the RCD, ideally with an impact hammer. If no impact hammer is available, then a small, lightweight hammer (without rubber attachment) can be used. The RCD should be hit only very lightly. Ideally, the RCD is hit on the tool. It is also possible for the tool end of the RCD to be hit. The RCD does not rotate while hammer blows are being applied.
   - Hammer blow to the RCD in X-direction (see marking on the RCD)
   - Hammer blow to the RCD in Y-direction (see marking on the RCD)
   - Hammer blow to the RCD in Z-direction (see marking on the RCD)

For the impact test, it must be ensured that the RCD is struck only once. Multiple contact during the attenuation process falsifies the signals. A separate measurement is to be initiated for each impact direction.
The sampling rate should be 10 000 Hz.

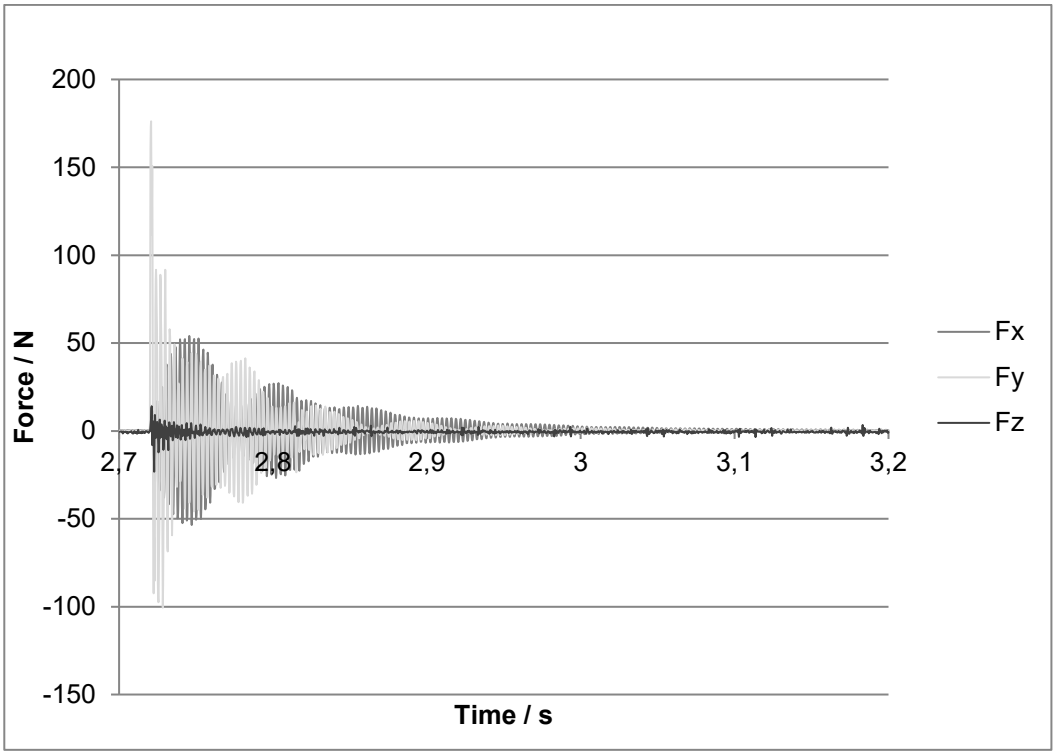


Fig. 25: Typical signal for connection in y-direction

Fast-Fourier-Transformation (FFT). With the aid of the FFT, the signals are converted to the frequency range; this is necessary in order for it to be at all possible to determine which interference frequencies are to be avoided. The so-called frequency response results from the FFT. An FFT is offered by many data acquisition programs, as for example the PTS App. The FFT is to be carried out for all three measurements.

6.4.2 Determination of the tooth passing frequencies to be avoided

The determination of the tooth passing frequencies to be avoided is accomplished with graphs. First, the tooth passing frequency must be determined.

$$f_{tooth\ passing} = \frac{n}{60} \cdot N$$

n: Speed [min^{-1}]
N: Number of cutting edges of the milling cutter

Afterwards, the frequency responses that were calculated with the help of the FFT are evaluated using graphs. All ranges in which a significant amplitude peak is visible are to be avoided. In the following example, these are the ranges 280 Hz … 360 Hz and 960 Hz … 1000 Hz. The speed of the tool must thus be selected in such a way that the tooth passing frequencies do not fall in these ranges.

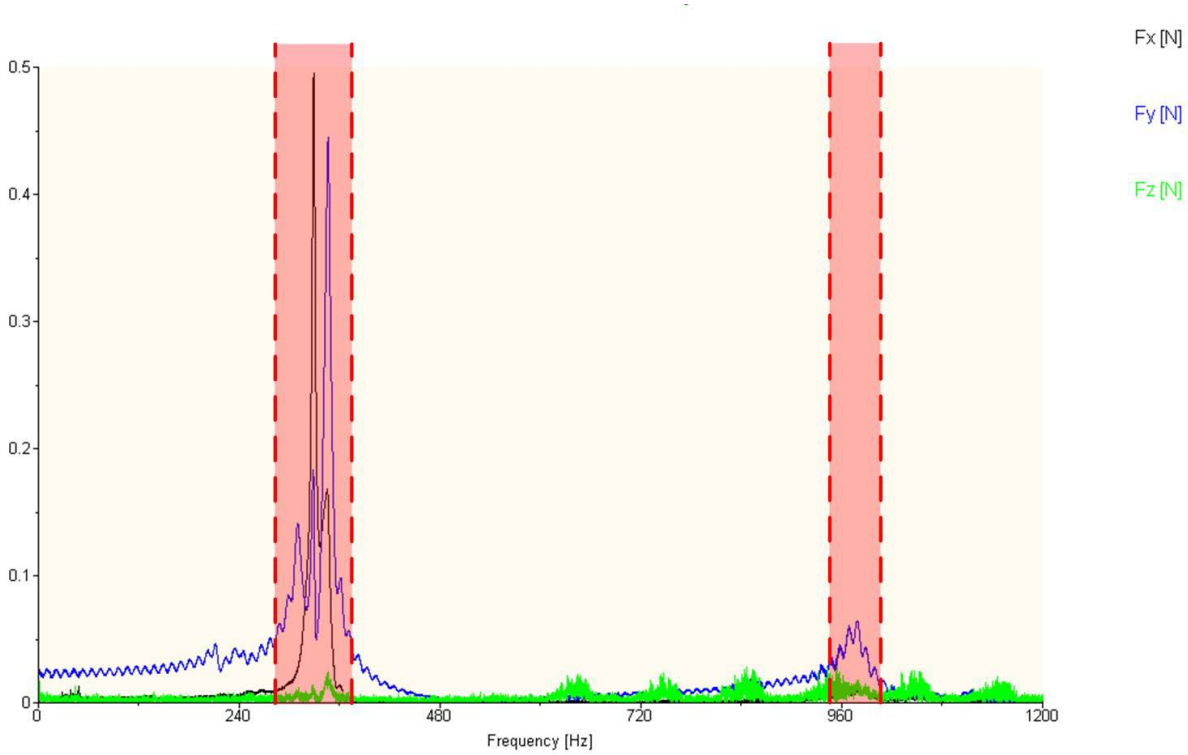


Fig. 26: Frequency ranges to be avoided

6.5 Signal conditioning

The measuring signals are pre-processed in the measuring system as well as in the receiver by a two-stage signal conditioning. The first stage serves to ensure the Nyquist-Shannon sampling rate and a measuring system-dependent digital filtering. This stage is predefined and cannot be influenced by the user. In the second stage, the user can set an individual low-pass filter whose cut-off frequency is at the cut-off frequency resulting from stage 1.

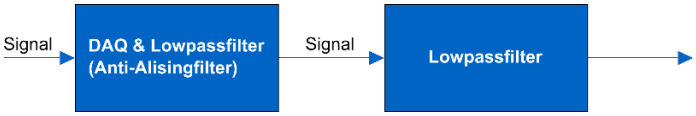


Fig. 27: Signal filtering procedure

6.5.1 Sampling rate and aliasing filter

With the RCD, one of up to three different sampling rates can be selected, depending on the expansion stage:

- 2.5 kHz
- 5 kHz
- 10 kHz

The aliasing filter (low-pass filter) serves to limit the bandwidth of a signal and thus to fulfil the Nyquist-Shannon sampling theorem. The following shows the specified filter behaviour of the different sampling rates for the RCD 9170B.

- 2.5 kS/Kanal

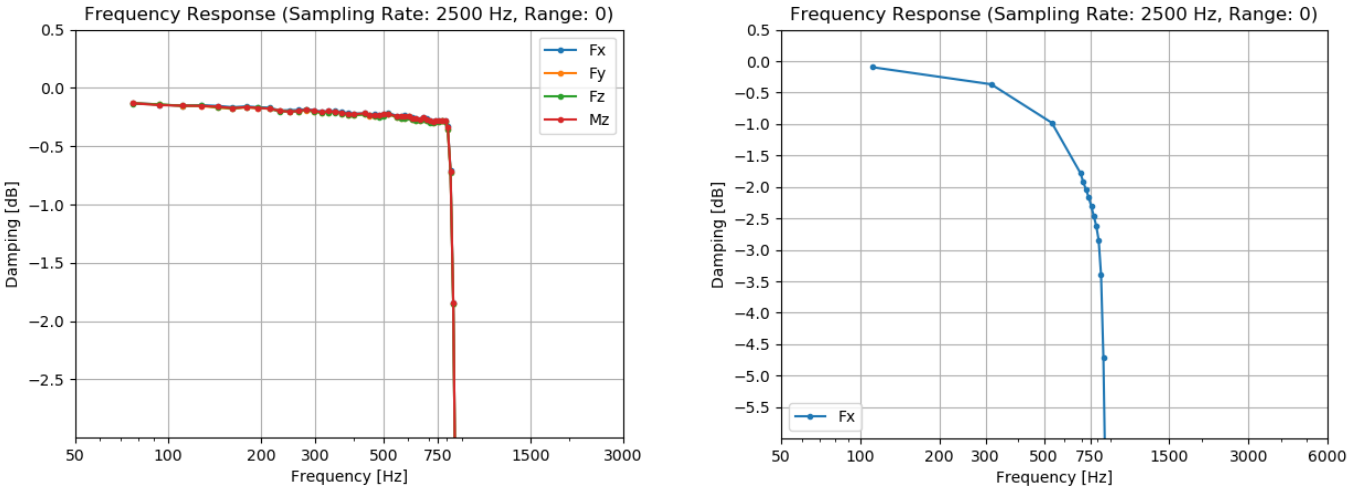


Fig. 28: Frequency response at 2.5 kS (left digital, right analogue)

- 5 kS/Kanal

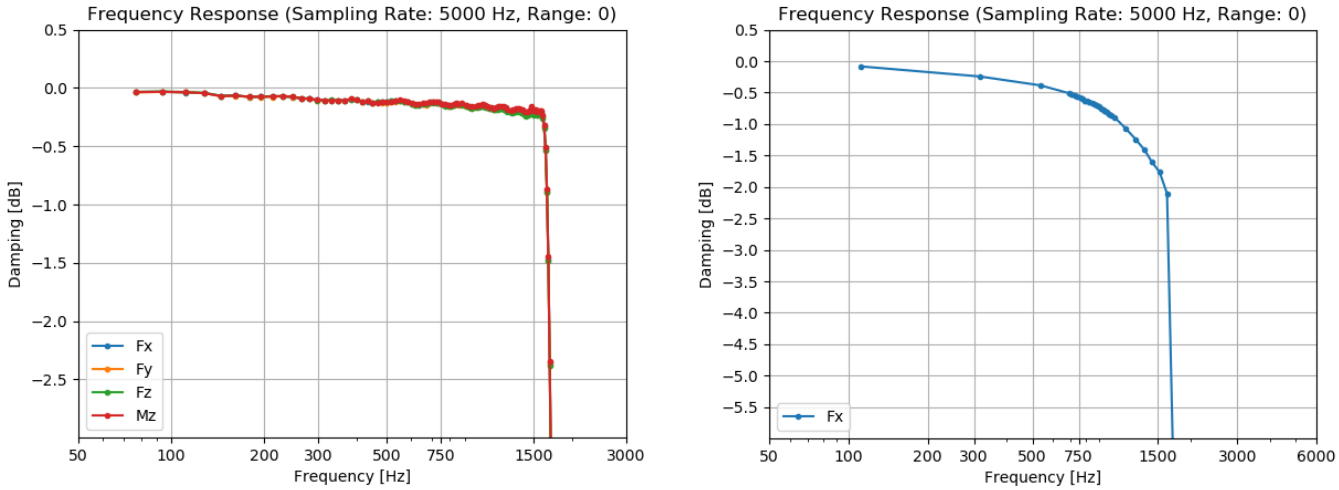


Fig. 29: Frequency response at 5 kS (left digital, right analogue)

- 10 kS/ Kanal

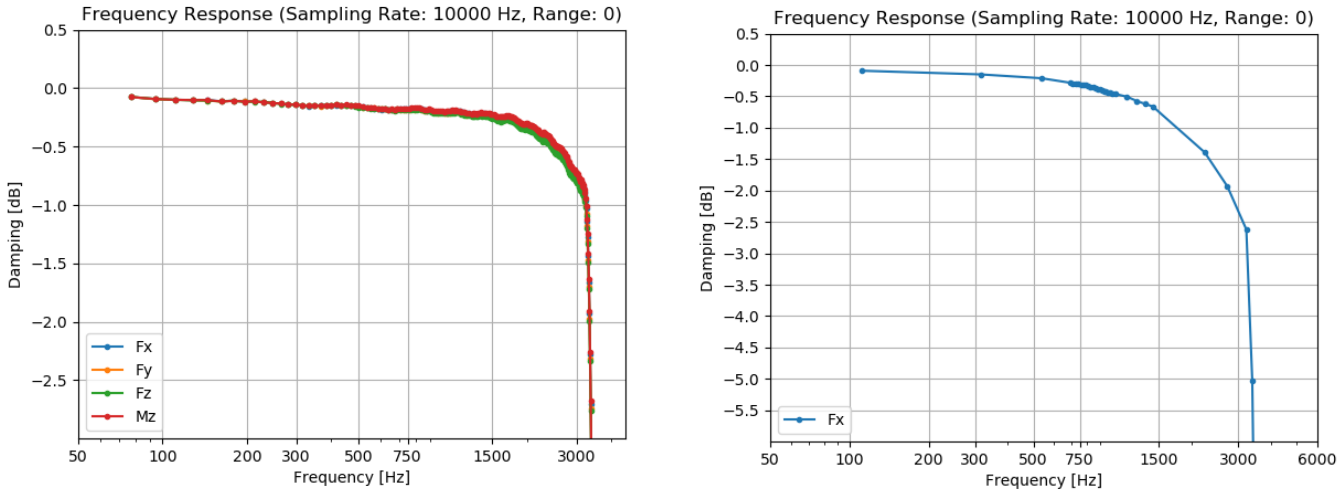


Fig. 30: Frequency response at 10 kS (left digital, right analogue)

6.6 Low-pass filter

In order to suppress any interfering signals, a further low-pass filter can be switched on by the user in the hardware setup of the software. The selectable frequencies are checked for consistency by the unit's firmware and are in the range below the cut-off frequency resulting from stage 1.

6.7 Sequence of a measurement

Once all of the settings mentioned in chapter 4 have been made, the measurement system is ready for operation and the tool can be inserted in the tool adapter. Various effects influence the quality of the measurement signals (see chapter 7). One should proceed as described in the following in order to minimize these to a large extent.

6.7.1 Procedure

1. **Select the measuring range:** Depending on the load calculated in chapter 5.2, the corresponding range is selected. If no calculation has been made and the loads are therefore unknown, it is recommended to keep the cutting parameters low for the first cut. By gradually increasing the cutting parameters, the user can approach the load limits.
2. **Start the spindle**
3. **Start the coolant supply**
4. **Start measurement** (set operate mode)
5. **Start feed rate.** The tool engages
6. At the end of the cutting process: **stop the feed rate**
7. **Stop measurement** (set reset mode)
8. **Stop coolant supply**
9. **Stop spindle**



The maximum speed – engraved on the rotor or specified in these instructions in chapter 10 – may not be exceeded.



Excessive effects of mechanical strains, in particular vibrations during machining, can damage the device. Chatter marks on the workpiece are clear signs of excessive vibration.



To activate the sensor, the charging cable must be connected for at least 3 seconds.

7. Influences on the measurement signals

The measurement signals are impaired by a variety of influences. The objective is to minimize these influences in order to obtain measurement results that are as unambiguous and informative as possible.

7.1 Influence of the speed on the signals

The centrifugal forces that affect the sensor increase by the square of the spindle speed. With Rotating dynamometers, the main effect of this load is on the F_z signal, while the two radial forces F_x and F_y are hardly impaired. The reason for this is the influence of centrifugal force on the crystal structure in the piezo element. A typical behavior of the F_z signal with increasing speed is indicated in the following.

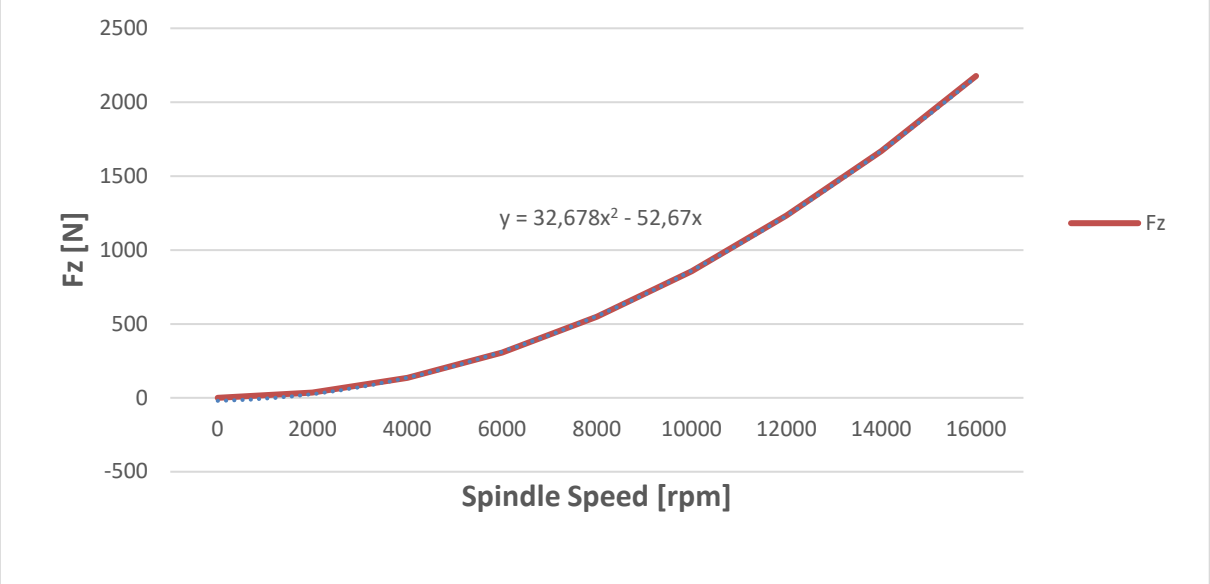


Fig. 31: Speed dependence of the F_z signal

7.1.1 Minimization of the influence of the speed

If the spindle is accelerated to the desired speed before the measurement is started ("Operate"), then the influence of the speed can be eliminated. The speed should be kept constant for the duration of the measurement.

7.2 Influence of gravitation on the signals

Gravitation has no influence on the signals for as long as the rotation axis of the rotor and thus also of the spindle is vertical. If the vertical position is abandoned, then gravitation will have an effect on all force signals.

If the rotation axis is horizontal, then only the two radial forces F_x and F_y will be influenced by gravitation. As a result of the rotation, the actual process signal will be superimposed by a sinus-shaped signal.

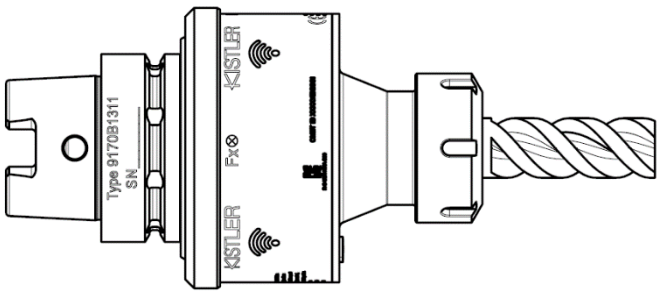


Fig. 32: Installation with horizontal rotation axis

In addition, the time of the "Operate" signal also has an influence. Two examples are shown in Fig. 33 and 34.

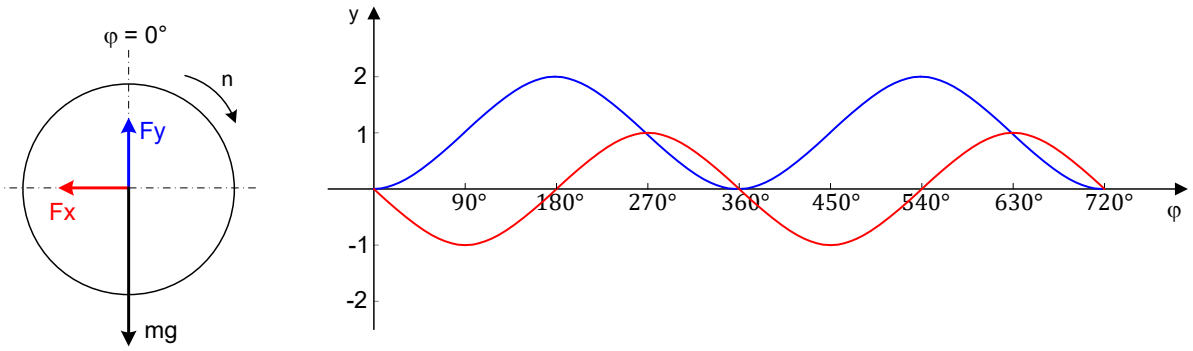


Fig. 33: Signals of the radial forces F_x and F_y with horizontal rotation axis when "Operate" takes place at φ = 0° and F_y aligned vertically upwards

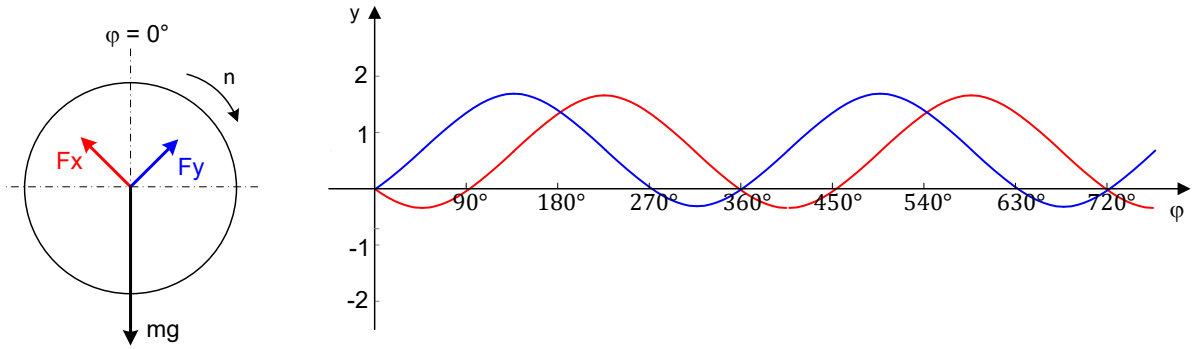


Fig. 34: Signals of the radial forces F_x and F_y with horizontal rotation axis when "Operate" takes place at φ = 0° and F_x and F_y are aligned upwards at an angle (45°)

The compensation for the influences of gravitation is difficult, particularly when the position of the rotor in space or the speed of the spindle changes during the period of the measurement.

7.3 Influence of the temperature on the signals

If temperature differences have an effect on the measurement system, then the measurement signal will also be influenced. The reaction of the z direction is particularly sensitive to temperature differences while the other directions are considerably more docile. This behavior has to do with the structure of the rotor.
The influence expresses itself in that the signal drifts away, even though no external change in terms of load is present. This drift due to temperature influences is usually non-linear.

Various factors can lead to temperature differentials:
- Change of location of the measuring device prior to the measurement application
- Heat induced by the coils in rotor and stator
- Friction heat in the bearings of the machine spindle
- Coolant lubricant
- Heat caused by the actual cutting procedure
- Local heating caused by prolonged gripping of the rotor

7.3.1 Minimization of the influence of temperature

In order to minimize the influence of temperature as much as possible, it is important that the system is thermally stabilized before the start of the measurement. This is accomplished through the following steps:
- Allow the measuring device to acclimatize itself to the utilization site for several hours before the measurement
- Switch on the measurement system at least 30 minutes before the measurement
- Allow the machine spindle with built-in Rotating dynamometer to warm up at the speed at which measurement will be carried out afterwards
- Allow coolant lubricant to flow through the rotor for a few minutes if internal coolant is used

7.4 Influence of internal coolant on the signals

NOTE

We recommend the use of the correct clamping nuts and sealing disks if the internal coolant is to be used

If the internal coolant is used, then the rotor will, on the one hand, not only feel the load caused by temperature changes on the one hand (see chapter 7.3) but also by the prevailing pressure. On the other hand, the axial direction of the rotor (z direction) is the one that is most affected.

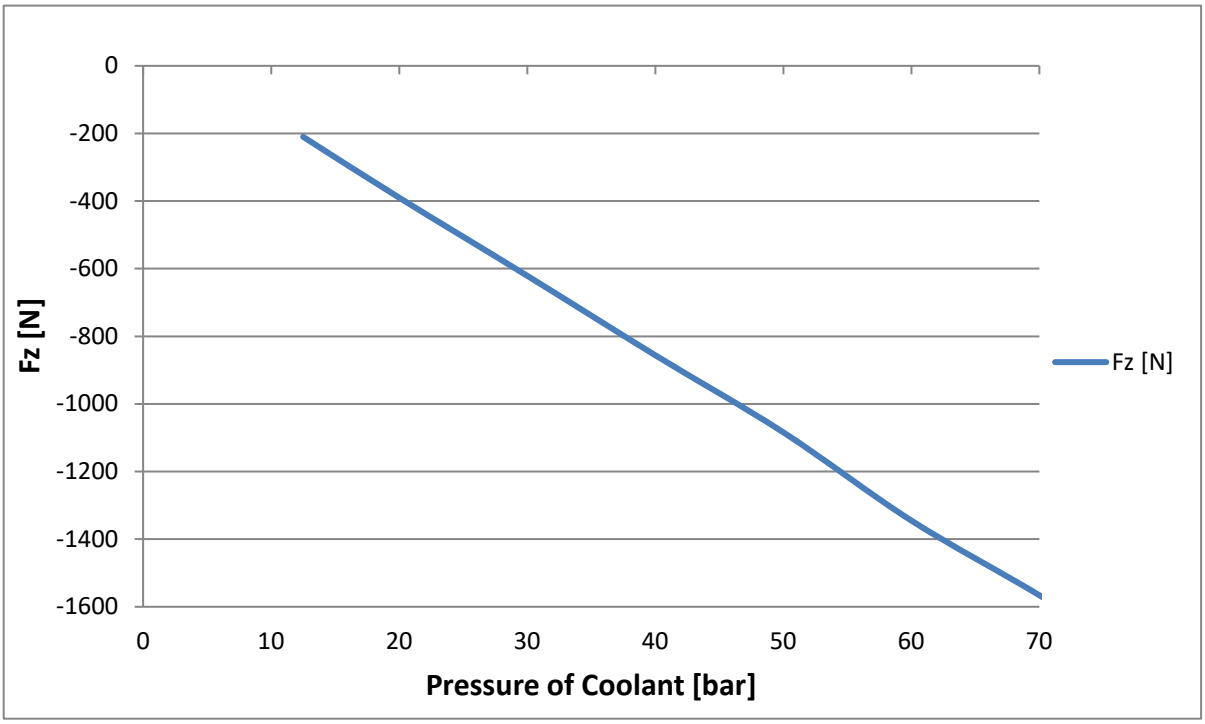


Fig. 35: Coolant pressure dependency of F_z

CAUTION

The maximum pressure of the internal coolant may not exceed 70 bar

7.4.1 Minimization ofthe Influence of internal coolant

If the internal coolant is started before the measurement is started ("Operate"), then the influence of the pressure on the internal coolant can be minimized. The pressure of the internal coolant is to be kept constant for the duration of the measurement.

8. Measurement signals

8.1 Interpretation of measurement signals



When interpreting the measurement signals, it must be noted that the coordinate system also rotates with the tool.

Thus it can be imagined that the user is sitting on the Rotating tool cutting edge and also experiencing the forces in effect during the cutting process.

8.1.1 Comparison of forces

As a basic rule, the measurement signals of a Rotating dynamometer do not directly read off the feed force F_f and the normal feed force F_{fN} but rather only calculate with the help of the immersion angle "φ" (see chapter 5.1).
.

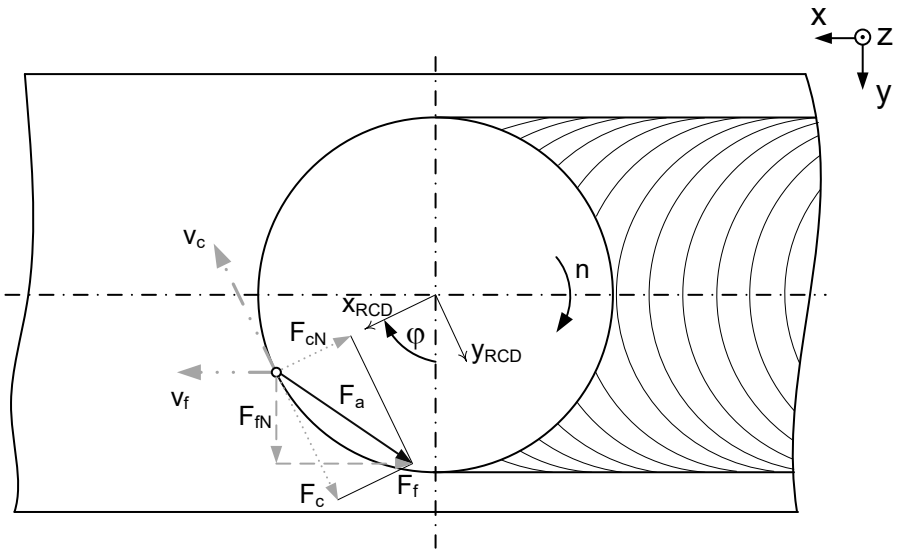


Fig. 36: Force vectors while face milling

One exception is the position at which the coordinate system of the rotating dynamometer matches the fixed coordinate system (φ = 90°).

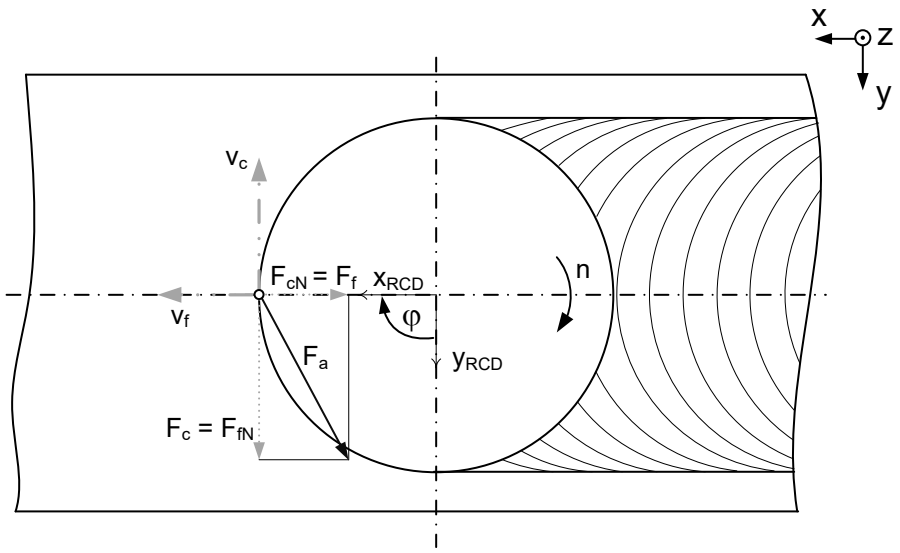


Fig. 37: Force vectors for φ = 90°

Assuming that the x axis RCD (x_{RCD}) is aligned precisely to the cutting edge, the following applies:

$$|F_c(\varphi = 90°)| = |F_{y,RCD}(\varphi = 90°)| = |F_{fN}(\varphi = 90°)|$$

$$|F_{cN}(\varphi = 90°)| = |F_{x,RCD}(\varphi = 90°)| = |F_f(\varphi = 90°)|$$

9. Servicing

9.1 Cleaning and storage

Keep the measurement electronics clean and protect the measuring chain against excessive dirt and moisture. In the event that the devices are not being used, we recommend storing them in the case supplied.

In case the device was affected by dirt, clean the device. Use only a clean, soft, lint-free cloth for cleaning. If necessary, moisten the cloth with water or pure alcohol. Do not use other liquids; these may attack the polymer components.

Only the machine and tool adapters should be treated with an oil spray or similar after use to protect against corrosion.

9.2 Recalibration of the devices

Recalibration of the dynamometer is necessary after an uncontrolled overload (see chapter 10), for example. This recalibration should be performed at the manufacturer's plant. Please contact the responsible Kistler representative in such cases.

Our products can be recalibrated or tested with the same procedure as was used prior to the initial delivery. All records are kept and are available to our customers so that the calibrations and repairs of our products can be tracked in their entirety.

Quality management systems, e.g. ISO 9001, require documentation of and compliance with periodic recalibrations of the measuring devices.

Kistler recommends this type of recalibration every two years.

9.3 Inspection for visible damage

Please observe the following instructions for the servicing of your dynamometer:

- Check the dynamometer for visible damage before each use. Do not put a damaged device into operation!
- Leave all repairs to the manufacturer's plant. Instructions for proceeding in such cases can be found in chapter 9.4 of this Instruction Manual



Important!

The spindle interface and the tool interfaces of the dynamometer are ground within very narrow tolerances. These interfaces are not permitted to be reworked!

9.4 In case of a defective dynamometer

If your dynamometer is defective, please observe the following instructions:

- Contact the Kistler representative responsible and give notice that the defective instrument is to be sent to you.
- Send the defective dynamometer in its original packaging to the Kistler representative.
- Enclose a precise description of the defect and the accompanying circumstances with the dynamometer.
- Describe the measuring process during which the defect occurred.
- If a major repair is necessary, you will receive a cost estimate.
- Kistler will it's best to repair your dynamometer in a short time and at minimum cost and send it to you in as-new condition.

10. Technical data

10.1 Dynamometer Type 9170B (Rotor)

Speed, max.		1/min	≤16 000
Measurement range nominal Single components, loads not simultaneous in their effects	F_x, F_y F_z M_z	kN kN N·m	–5 … 5 –20 … 20 –210 … 210
Measuring Range 1	F_x, F_y F_z M_z	kN kN N·m	–0.5 … 0.5 –2.5 … 2.5 –10 … 10
Measuring Range 2	F_x, F_y F_z M_z	kN kN N·m	–1 … 1 –5 … 5 –20 … 20
Measuring Range 3	F_x, F_y F_z M_z	kN kN N·m	–2.5 … 2.5 –10 … 10 –50 … 50
Measuring Range 4	F_x, F_y F_z M_z	kN kN N·m	–5 … 5 –20 … 20 –100 … 100
Calibration range in the function of the spindle adapter			see chapter 10.1.1
Sensitivity range 1	F_x, F_y F_z M_z	mV/N mV/N mV/N·m	≈20 ≈4 ≈1000
Sensitivity range 2	F_x, F_y F_z M_z	mV/N mV/N mV/N·m	≈10 ≈2 ≈500
Sensitivity range 3	F_x, F_y F_z M_z	mV/N mV/N mV/N·m	≈4 ≈1 ≈200
Sensitivity range 4	F_x, F_y F_z M_z	mV/N mV/N mV/N·m	≈2 ≈0.5 ≈100
Linearity		%FSO	≤±1
Hysteresis		%FSO	≤1
Crosstalk	F_x <–> F_y F_z –> $F_{x,y}$ $F_{x,y}$ –> F_z M_z –> F_z F_z –> M_z	% FSO % FSO % FSO N/N·m mN·m/N	≤±2 ≤±1 ≤±3 ≤±1 ≤±1
Natural frequency Type 9170B131x (with integrated spindle adapter HSK-A63) measured without tool, collect chuck and clamping nut	$f_{n,x}$ $f_{n,y}$ $f_{n,z}$	Hz Hz Hz	≈2 000 ≈2 000 ≈5 300

Sampling rate per channel [1]		kHz	2.5 / 5 / 10
Bandwidth		kHz	≈ 0 … 0.8 / 1.6 / 3.2
Resolution		bit	16
Battery life (active measurement)		h	> 5
Supply voltage (charging)		V	5
Battery charging time		h	< 4
Max. storage time until recharge		months	3
Transmission type			GFSK
Frequency range (data transmission)		MHz	2400 … 2480
Max. transmission power		dBm	+8
Range [2]		m	≤5
Operating temperature		°C	0 ... 60
Charging temperature		°C	0 … 45
Storage temperature		°C	–20 … 45
Degree of protection (IEC 60529)			IP67
Degree of pollution			2
Internal cutting fluid pressure, max.		bar	≤70
Balancing class		G	≤2.5
Weight Type 9170B131x (rotor only) (with integrated spindle adapter HSK-A63) Measured without tool, collect chuck and clamping nut		kg	≈2

[1] Selectable sampling rate per channel (not separately selectable)
[2] External influencing factors (such as objects, other radio waves) can affect the range and connection stability.

10.1.1 Calibration ranges in function of the spindle adapter

Particularly in the case of smaller machine adapters, this limits the useful range of the rotating dynamometer. In the case of planar-fitting spindle interfaces (HSK and Capto adapters), the maximum bending moment may not exceed the limit value so that the rotor does not tip over and lose the planar contact. The rotors are calibrated in accordance with these limit values (see chapter 5.2.2).

			Calibration range 1	**Calibration range 2**	**Calibration range 3**	**Calibration range 4**
9170B111x	HSK-A40	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 500 10	1 000 3 000 20	0 6 000 40	0 12 000 80
9170B121x	HSK-A50	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	1 500 10 000 50	0 20 000 100
9170B131x	HSK-A63	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	2 000 10 000 50	3 000 20 000 100
9170B161x	HSK-E40	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 500 2,5	1 000 3 000 5	0 6 000 10	0 12 000 15
9170B171x	HSK-E50	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 5	1 000 5 000 10	1 500 10 000 20	0 20 000 35
9170B181x	HSK-E63	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	2 000 10 000 40	3 000 20 000 70
9170B211x	DIN ISO 7388-1 - AD30 (DIN 69871-AD30)	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 500 10	1 000 3 000 20	0 6 000 40	0 12 000 80
9170B221x	DIN ISO 7388-1 – AD40 (DIN 69871-AD40)	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	2 000 10 000 50	3 000 20 000 100
9170A241x	JIS B 6339-2 JD 30 (MAS 403 BT 30)	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 500 10	1 000 3 000 20	0 6 000 40	0 12 000 80
9170A251x	JIS B 6339-2 JD 40 (MAS 403 BT 40)	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	2 000 10 000 50	3 000 20 000 100
9170A271x	ANSI / ASME B5.50-30 (CAT 30)	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 500 10	1 000 3 000 20	0 6 000 40	0 12 000 80
9170A281x	ANSI / ASME B5.50-40 (CAT 40)	F_x,F_y [N] F_z [N] M_z [N·m]	500 2 500 10	1 000 5 000 20	2 000 10 000 50	3 000 20 000 100
9170A311x	Capto C5	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 000 10	1 000 2 000 20	1 500 4 000 30	0 8 000 60
9170A321x	Capto C6	F_x,F_y [N] F_z [N] M_z [N·m]	500 1 250 10	1 000 2 500 20	2 000 5 000 40	3 000 10 000 80

10.2 LabAmp WL-Receiver Typ 5347A4…

Number of channels			4
Number of ranges per channel			4
Digital low-pass filter [1] Cutoff-frequency (-3 dB) Selection in 1 Hz steps 1)		Hz	≥ 10
Scaling analogue output [2]		V	±10
Time shift analogue output		ms	320 … 400
Output impedance typically		Ω	10
Max. Output current		mA	±2
Signal output connector type			BNC neg.
Ethernet interface [3]			2 x RJ45
Remote control			D-Sub 9f
Supply voltage range		VDC	18 … 30
Power consumption PowerOff: StandBy: Reset: Measure:		mW	 < 0.4 < 11 < 170 < 180
Operating temperature range		°C	0 ... 60
Rel. humidity, not condensing		RHS %	≤ 90
Degree of protection (IEC 60529)			IP20
Outer dimensions incl. feet and connectors (WxHxD)		mm	218x50x223
Weight		kg	1.2

1) The PTS App offers selectable ranges
2) Corresponds to the selected full measuring range of the calibration
3) The Sync Out connection is not active

10.2.1 Time offset during transmission of the signals

To ensure continuous measurement data output at the analogue outputs of the receiver, a buffer was integrated in the receiver that can safely bridge a transmission interruption of the data from the sensor to the receiver of < 300 ms. As a result, the measurement data output at the analogue outputs takes place with a time delay.

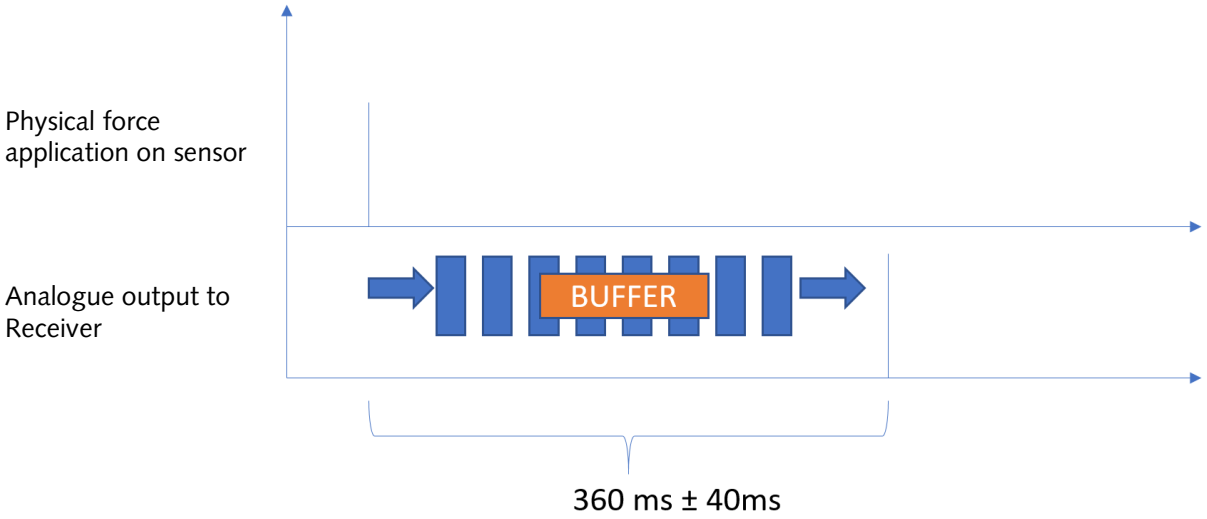
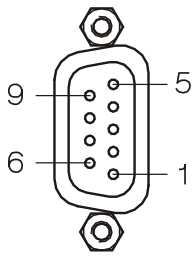


Fig. 38: Time offset analogue output

10.2.2 Remote Control

Remote Control



Pin 1	+24 VDC
Pin 2	EGND
Pin 3	---
Pin 4	/Trigger
Pin 5	/Measure & Reset
Pin 6	DGND
Pin 7	---
Pin 8	---
Pin 9	---

Remote input:
2 – 5: Operate and reset the charge amplifier
2 – 4: Trigger signal start - stop measurement. The event generated is interpreted depending on the software used. The PTS app also automatically sets the charge amplifier in the RCD to operate mode when the measurement is started and back to reset mode when the measurement is stopped.

(Digital input and 24V output voltage)
Remote /Measure and Remote /Trigger with internal pull-up 10kΩ to +5V

Connector type		D-Sub 9f
Input level		
High (Reset, Stopp Trigger)	V	>3.5
or		Input open
Low (Measure, Start Trigger)	V	<1
Max. input voltage	V	±30
Supply output	V DC	+24/±10%
Output current (short circuit proof)	mA	≤200

10.3 Receiver Dongle

Outer dimensions	LxWxH	mm	51x19x11
Weight		g	<10
Connector type			USB-A
Max. transmission power		dBm	+8
Frequency range		MHz	2400-2480

10.4 Components and accessories

10.4.1 Components

	Type/Art. No.
▪ Rotor	9170B
▪ Power supply* 5 V	55245931
▪ Country-specific plug set	55255315
▪ Magnetic charging cable to power supply	55245928
▪ WL Receiver	5347A
▪ Power supply* 24 V incl. country-specific plug	5779A2
▪ Ethernet cable (Cat. 6e) l = 2.0m	55117964
▪ Receiver dongle	55250534
▪ USB 2.0 A extension cable	55255530
▪ License dongle for PTS App	2935A
▪ Peli Case roller case	55245041

10.4.2 Included accessories

	Type/Art. No.
▪ Clamping wrench for ER collets	65007932
▪ Clamping nut Hi-Q/ER32 for ER collets	65007915

10.4.3 Optional accessories

	Type/Art. No.
▪ Collets DIN 6499-B32-UP	9169Axx
▪ USB 2.0 A extension cable	55255530

11. Annex

11.1 Glossary

Best straight line

see Annex 11.3.

Cable capacitance

The cable capacitance, and thus the length of the connecting cable, has no noteworthy influence on the measuring result when Kistler special cables and Kistler charge amplifiers are used.

Calibrated range

Measuring range or part of the measuring range for which the sensor has been calibrated.
Note: Because of the high linearity of quartz crystal sensors, the sensitivity of a measuring range can usually also be used for smaller measuring ranges.

Calibration certificate

Document for sensors and devices stating the results of the factory calibration.

Charge amplifier

Part of a measuring chain which converts the charge signal from the sensor into a proportional voltage signal or current signal.
Note: The linearity of charge amplifiers is typically within the range of ±0,01 % of the measuring range selected. For the accuracy of the measurement, this is usually negligible compared with other influence quantities.

Charge signal

Physical quantity which a piezoelectric sensor yields on its mechanical load.

Coulomb

Unit of electric charge.
1 coulomb corresponds to 1 Ampere-second (1 C = 1 As).

Crosstalk

Signal at the output of a sensor, produced by a measurand acting on the sensor, which is different from the measurand assigned to this output. For example, when a load in the F_y direction produces an F_z signal in a three-component sensor. In terms of electrical devices, it is a measure for the signal impact acting from a channel to the neighboring ones.

Degree of protection

Protection of electrical equipment by suitable enclosures, covers etc. according to EN60529. The protection class is stated by IP (International Protection) followed by two digits. The first digit stands for protection against touching and the ingress of solid bodies, the second for the degree of protection against the ingress of water.
Example: IP65 indicates a complete protection against touching, as well as against the ingress of dust and spray water from all directions.

Disturbance

Quantity that is not the measurand but that affects the result of the measurement. The contributions are expressed in terms of the measurand related to the acting quantity.
Examples:
additional bending moment acts on a force sensor:
$M_{x,y} \rightarrow F_z$
additional axial force acts on a torque sensor:
$F_z \rightarrow M_z$
additional shear force acts on a force/torque sensor:
$F_{x,y} \rightarrow F_z$ and $F_{x,y} \rightarrow M_z$

Drift

Unwanted changes in the output signal independent of the measurand as a function of time.

Frequency range

see Annex 11.4.

FSO

Full Scale Output. Range of measurement signal of a sensor, corresponding to the full scale of the measurand. It is the algebraic difference of the upper and lower limit of the measurement signal range.
Example: A pressure sensor having the measuring range from 0 bar to 2 bar and the corresponding range of measurement signal from –5 … +5 V presents the FSO: 10 V. An electrical monitoring system having an input range from
0 … 10 V and the corresponding output range from 4 … 20 mA shows the FSO: 16 mA

Ground insulation

High electrical resistance of a sensor between signal line and ground, or of a charge amplifier between connector screen and ground.

Hysteresis

see Annex 11.3.

Insulation resistance

Electric resistance of a sensor, cable or the input of a charge amplifier measured between the signal line and the connection ground (sensor body), while the test voltage is stated accordingly. The insulation resistance applies for piezoelectric sensor, strain gauge sensors and semi-conductor sensors

Linearity

see Annex 11.3.

Measurand

Physical quantity, state or characteristic which is measured, e.g. force, torque, pressure etc.

Measuring chain

Interconnection of several individual components to meet measuring requirements. Measuring chains usually consist of sensors and amplifiers in conjunction with data acquisition, display, evaluation and recording equipment (e.g. PC, printer).

Measuring range, charge amplifier

Charge, voltage or current range of the signal input. Entry in units of the measurand is also possible depending on the type of charge amplifier.

Measuring range, sensor

Range in which the quality of the measurement within the stated tolerances is guaranteed. This range must be regarded as a binding maximum range.

Natural frequency

Frequency of free (not forced) oscillations of the entire sensor. In practice the (usually lower) natural frequency of the entire mounting structure governs the frequency behaviour.

Operating temperature range

Range of ambient temperatures in which the sensor is to be operated. The temperature-dependent tolerances stated apply only within this range.

Output signal

see "Charge signal"

Overload

Maximum value of the measurand with which a sensor can be loaded without sustaining damage. This refers to a safety margin and is not an extended measuring range. The characteristics specified in the calibration certificate are no longer guaranteed in the event of an overload. Nevertheless, measurements made during an overload in most cases provide useful results

pC (picocoulomb)

1 picocoulomb = 10^{-12} coulomb. See "Coulomb".

piezoelectric

Characteristic of crystals (e.g. quartz) in which mechanical loading produces a proportional electrical charge.

Pretension

Continuous force loading of a measuring element resulting from the application of tension sleeves or tension bolts so that an applied force (force or pressure sensors) is acting on the measuring element as well as on the tension element (force shunt).

Preload

Loading of a calibration object prior to the calibration cycles for realization of a well-defined load at the beginning and end of the calibration cycles.

Prestress

Repeated loading (1 … 3 times) of a calibration object prior to the calibration cycles applied for eliminating stresses within the object and/or for ensuring a rigid force application.

quasistatic

Describes the ability of Kistler sensors and charge amplifiers to undertake short-term measurements or DC-similar measurements.

Range

see "Measuring range"

Scaling

Output voltage per unit of the measurand at the analogue or monitor output of a charge amplifier.

Sensitivity

Nominal value or calibrated value stated in the calibration certificate of the change in the response of a sensor divided by the corresponding change in the value of the measurand.
Note: sensitivity of piezoresistive and strain gauge sensors is additionally dependent on the excitation current or voltage.

Sensor

System which produces a definite change in the output signal as a function of the change of the measurand acting on it.
Note: The term "Sensor" is equivalent to "Transducer"

Temperature influence

see Annex 11.5.

Threshold

Smallest change in the measurand that produces a measurable change in the sensor output, while the change of the measurand takes place slowly and monotonically.
Note: In practice, the rule of thumb applies that the threshold is about two to three times as large as the typical noise signal of a charge amplifier. This value can, however, only be achieved in dynamic measurements, whereas with quasi-static measurements, drift and environmental influences are limiting factors

Time constant

The time constant describes the behavior of a high-pass filter and represents the time after which the signal is reduced to 1/e of the output value.
Note: The time constant enables the measuring error to be estimated in relation to the measuring duration. You will find detailed information on time constants and sensitivity ranges in the operating instructions for your charge amplifier.
Example: The time constant depends on the measuring range selected on the charge amplifier. Possible values vary from approx. 0,01 s in the most sensitive range to approx. 100 000 s in the least sensitive range. The largest possible time constant must be selected for quasi-static measurements.

11.2 Measurement Uncertainty

Systematic errors, accuracy
Accuracy is the extent of the conformity between a measured value and a true value of the measurand. In a piezoelectric measuring chain it is determined by many systematic errors, such as

- Sensor linearity
- Sensor hysteresis
- Crosstalk from other measurands
- Charge amplifier linearity
- Disturbance (forces, moments)
- Disturbance (environmental influences, like temperature)
- Duration of the measurement
- etc.

Experience with a measuring chain consisting of sensor, cable and charge amplifier shows that an accuracy in the range of 1 ... 2 % of the measuring range is achieved. This value does not include errors due to influences from external sources acting on the measuring chain, due to mechanical adaptation of the sensor and environmental influences. For the highest accuracy requirements, we recommend calibration of measuring ranges specific to the application.

Measurement uncertainty of charge amplifiers
When random and systematic errors together are quantified as variance, measurement uncertainty can be derived. With charge amplifiers, this mainly depends on type. The following typical values apply:

Laboratory charge amplifier	±0.2 ... 0.5 %FSO
Industrial charge amplifier	±1 %FSO

Higher accuracy can be achieved with the following procedures:

- Calibration in the Kistler factory
- Calibration with charge amplifier Type 5395A
- Restriction of the temperature range

Random errors, precision, reproducibility
Precision or reproducibility is the extent of conformity between independent data measured under specified conditions.

Repeatability
Repeatability is understood as "serial precision" for example conformity between several measurements in sequence under largely unchanged conditions.

This requirement is found mainly in repetitive measurements in production processes, where good repeatability is usually sufficient for process monitoring. Accuracy primarily plays only a subordinate role, when measurements can be directly related to conforming/ nonconforming parts.

For good repeatability, piezoelectric measuring technology offers the particular advantage that the charge can be discharged with <Reset> before every measurement, enabling the zero point to be re-determined. Errors due to zero drift caused by influences changing with time, such as the temperature, are thus basically excluded.

With Kistler piezoelectric measuring chains, a typical repeatability within 0,1 %FSO can be assumed.

11.3 Linearity

Sensor linearity
The quartz crystal produces an electric charge, which is exactly proportional to the load. However, certain unavoidable deviations occur due to the mechanical construction of the sensor. Linearity represents the maximum deviation between ideal and actual output signal characteristics in relation to the measurand in a specific measuring range. It is expressed in percentage of the particular measuring range limit and is defined according to ANSI/ISA-S37.1 as the closeness of the calibration curve to a "best straight line" passing through the zero point:

"Best Straight Line" – A line midway between the two parallel straight lines closest together and enclosing all Output vs. Measurand values on a Calibration Curve.

The best straight line can be determined as follows:

Best straight line – geometric definition
A line midway the two parallel straight lines closest together and enclosing all output versus measurand values on a calibration curve. In addition, it must pass through the zero point based on the assumption that zero measurand results in zero output signal.
Note: The slope of this center line corresponds to the sensitivity of the sensor. Half the distance between the two parallels (measured in the ordinate direction) is the linearity

Best straight line – mathematical definition

- The minimization of maximum deviation is known as Chebyshev's approximation. The best straight line is determined as follows:
- x = measurand (reference)
- Q = sensor charge signal or output signal from the charge amplifier
- $Q(x)$ = calibration curve, rising and falling
- s = slope of the best straight lines
- Best straight line: $y_i = s\ x_i$ (with starting value for slope s)
- Form residues: $res_i = Q_i - y_i$
- $\Sigma_{res} = res_{max} + res_{min}$ sum min. + max. deviation
- Recursive minimization of $\Sigma_{res} = f(s)$ by changing s until $\Sigma_{res} = 0$
- Linearity $a = res_{max} = |res_{min}|$

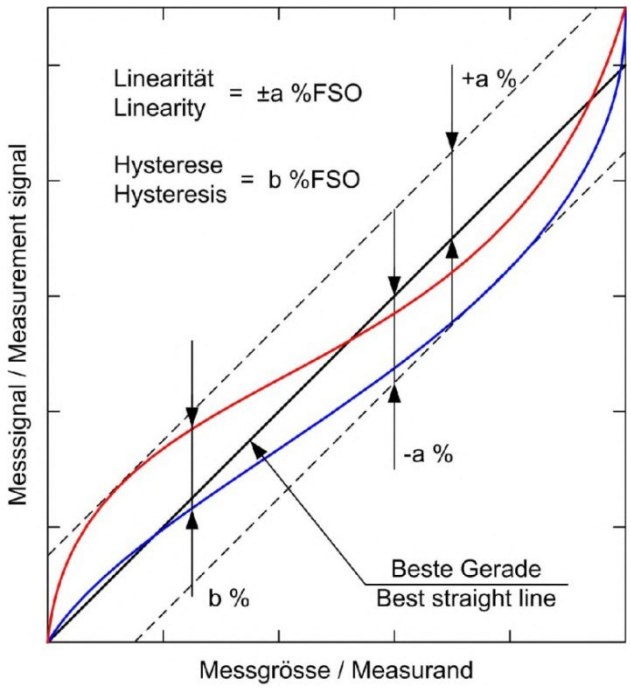


Fig. 39: Best straight line, linearity and hysteresis

Hysteresis

The maximum difference in output, at any measurand value within the specified range, when the value is approached first increasing and then decreasing measurand (source: ANSI/ISA-S37.1).

Note: The quartz crystal itself has a scarcely measurable hysteresis. However, the mechanical construction of the sensor can result in slight hysteresis. If the hysteresis is above the specified values (in %FSO), then the sensor is faulty or has not been correctly installed.

Charge amplifier linearity

The linearity of charge amplifiers is typically within the range of ±0,05 % of the measuring range selected. For the accuracy of the measurement, this is usually negligible compared with other influences.

11.4 Frequency Range

Because of their mechanical quality, piezoelectric sensors have very low damping. The useful frequency range is limited in the upwards direction by the increasing resonance rise.

Key: **f = Measuring frequency**
f_n = Natural frequency
A/A_0 = Amplitude ratio

The following approximate values apply to the amplitude error or achievable accuracy as a function of frequency:

Accuracy 10 % → f_{max} ≈ 0,3 · natural frequency
5 % → f_{max} ≈ 0,2 · natural frequency
1 % → f_{max} ≈ 0,1 · natural frequency

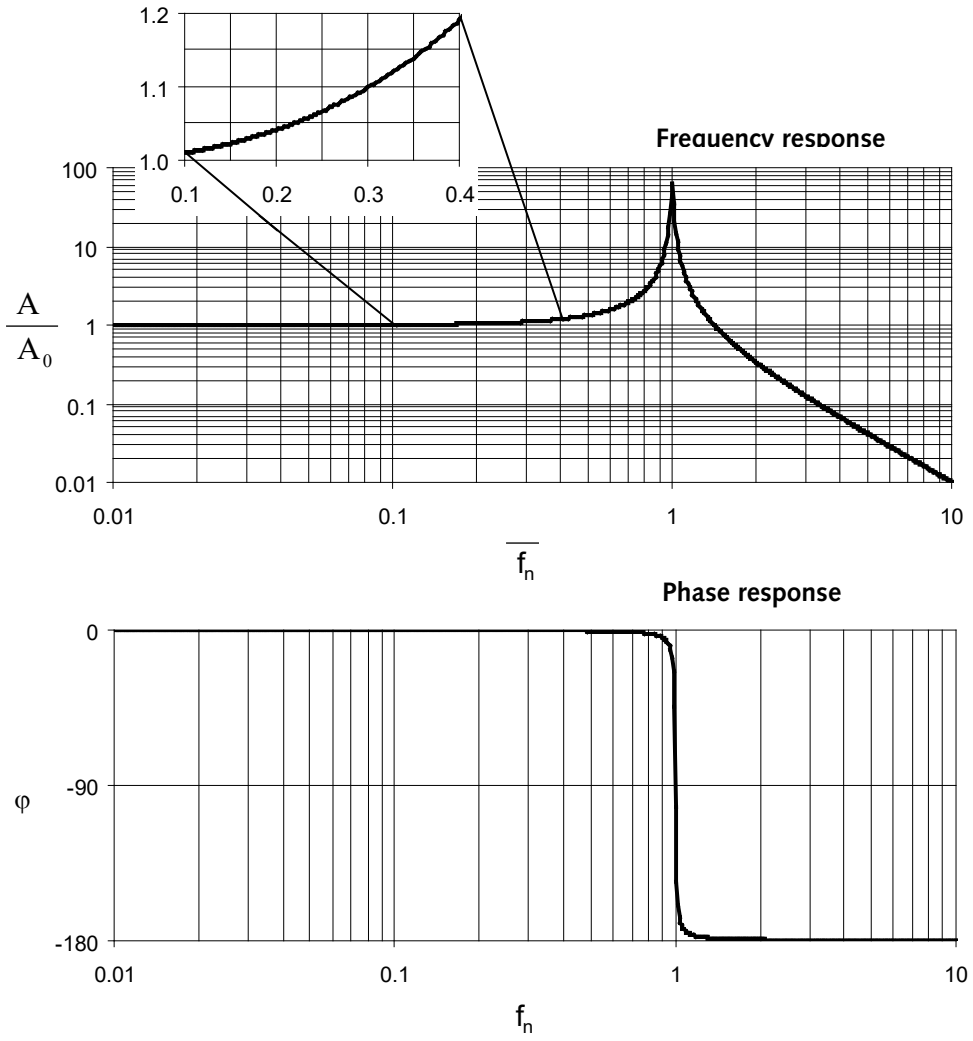


Fig. 40: Schematic representation of frequency response and phase response

In their dynamic behavior, piezoelectric sensors are superior to all other measuring methods. Their high rigidity results in the highest possible natural frequencies. Piezoelectric sensors are thus ideal for measuring measurands which change rapidly over time. Their dynamic behavior is thereby largely determined by the surrounding structure. Therefore the frequency response of the entire measuring arrangement must be investigated for the largest possible, useful measuring range.

There are two possibilities here:

Frequency analysis
The measuring arrangement is stimulated e.g. with an impulse hammer and the sensor output signals then subjected to a frequency analysis. The results is presented in terms of an amplitude and phase response curve

FEM
Finite Element Method: Finite Element Method (FEM) is a numerical procedure for the approximate solution of differential equations having boundary conditions. Using the FE-Method, various problems from physical areas can be calculated. A computing field obtains an arbitrary number of elements, however, being finite in size and number.
Note: In the finite element method, a homogeneous body is substituted for the sensor closely approximating to the dimensions and average density of the sensor. The average modulus of elasticity of this equivalent body is then continuously varied until its natural frequency coincides with that of the actual sensor (technical data). An equivalent body defined in this way is usually a good approximation of the sensor. The equivalent substitute body is inserted into the structure to be simulated, and by this means the natural frequency of the structure is calculated. Using this procedure, the FEM can be used to determine the frequency behavior of a measuring arrangement with good approximation

11.5 Influence of Temperature

Temperature changes during a measurement result in an error signal in the form of a zero drift. In critical applications, we recommend that protection be provided for the sensor as far as possible against changes in temperature.

Temperature error of the zero point (static error)
Temperature error [unit of the measurand/°C] is the greatest change to the output signal in a specified measuring range after a specific sensor temperature change, following which the sensor is again in thermal equilibrium with its environment. Temperature errors are caused by changes in stress in the sensor, which in turn are influenced by the preload or installation conditions.

Temperature gradient error (dynamic error)
A temporary change in the output signal is denoted as temperature gradient error, when the temperature of the environment or surrounding medium changes with a certain rate. In this case, the sensor is not in thermal equilibrium with the environment.
The temperature gradient error is primarily determined by the installation conditions and the application, and cannot be generally specified. However, the temperature gradient error can be significant, particularly in the case of sensitive measurements and small measured values. It is therefore extremely important to keep the sensor temperature constant during the actual measuring time.

Immersion Bath Error 150 ... 200 °C

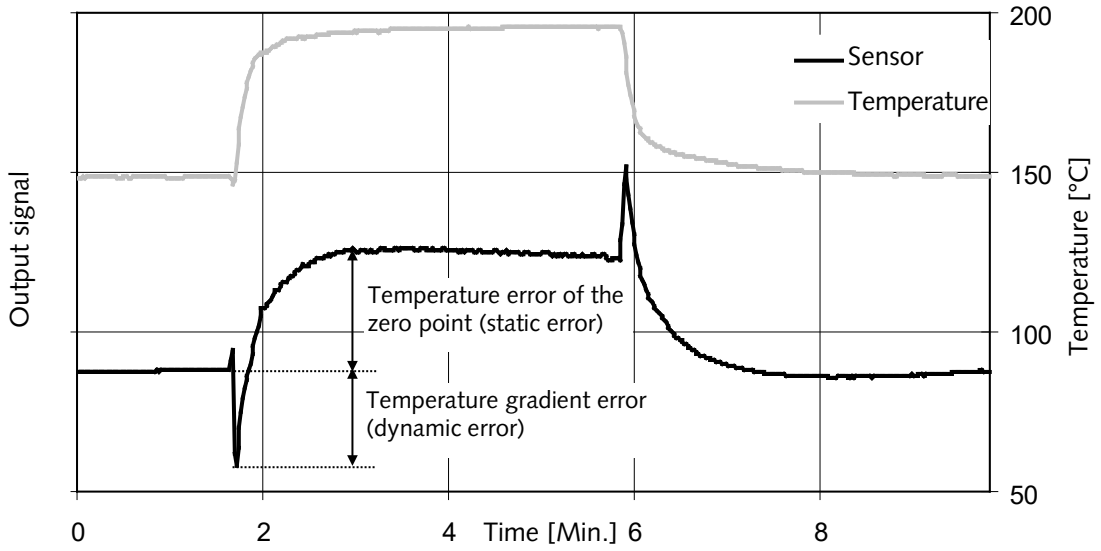


Fig. 41: Example of a temperature error of the output signal zero for a temperature rise from 150 °C to 200 °C (immersion bath).

Temperature coefficient of sensitivity
Change in the sensitivity, i.e. the slope of the best straight line, as a function of temperature. The temperature distribution in the sensor is assumed to be homogeneous, and in thermal equilibrium with the environment. The temperature coefficient of the sensitivity is typically only approx. 0,02 %/°C, and is thus mostly negligible compared with other influences

12. Declaration of conformity

12.1 CE (Europe)

Hereby, Kistler Instrumente AG declares that the radio equipment type 9170B... is in compliance with Directive 2014/53/EU.
The full text of the EU declaration of conformity is available at the following Internet address:
www.kistler.com > About us > Certificates and product compliance > Product compliance

12.2 UK

Hereby, Kistler Instrumente AG declares that the radio equipment type 9170B... is in compliance with Directive UK SI 2017 Nr. 1206.
The full text of the EU declaration of conformity is available at the following Internet address:
www.kistler.com > About us > Certificates and product compliance > Product compliance

12.3 FCC (USA)

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

12.4 IC (Kanada)

12.4.1 Canada Compliance Statement

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's license exempt RSS(s). Operation is subject to the following two conditions:

- This device may not cause interference
- This device must accept any interference, including interference that may cause undesired operation of the device

Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be chosen in such a way that the equivalent isotropically radiated power (e.i.r.p.) is not more than that is necessary for successful communication.
This equipment complies with IC RSS-102 radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20 cm between the radiator and your body.

12.4.2 Canada Déclaration de Conformité

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes:

- L'appareil ne doit pas produire de brouillage
- L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement

Conformément aux réglementations d'Industry Canada, cet émetteur radio ne peut fonctionner qu'à l'aide d'une antenne dont le type et le gain maximal (ou minimal) ont été approuvés pour cet émetteur par Industry Canada. Pour réduire le risque d'interférences avec d'autres utilisateurs , il faut choisir le type d'antenne et son gain de telle sorte que la puissance isotrope rayonnée équivalente (p.i.r.e) ne soit pas supérieure à celle requise pour obtenir une communication satisfaisante.

Cet équipement respecte les limites d'exposition aux rayonnements IC RSS-102 définies pour un environnement

non contrôlé. Il doit être installé et utilisé en maintenant une distance minimum de 20cm entre le radiateur et votre corps.

12.5 Japan

Japanese Radio Law and Japanese Telecommunications Business Law Compliance:

- This device is granted pursuant to the Japanese Radio Law (電波法) and the Japanese Telecommunications Business Law (電気通信事業法).
- This device should not be modified (otherwise the granted designation number will become invalid).

12.6 KC (South Korea)



There is a possibility of radio interference during operation of this radio facility.